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US Rates Vol Lab

Trading long-dated convexity

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- **Weekly Focus:** Convexity embedded in long-dated forward flatteners is typically cheap to realized vols. Systematic delta-hedging of long-dated flatteners has been historically profitable while also offering a long vol bias. We believe these trades are attractive now given valuations and our medium-term long view on vol. We especially like 15y5y/20y10y delta-hedged flatteners.
- **Vol Surface:** Upper-left volatilities remain relatively rich on the surface. Midcurves on front-end rates look especially rich.
- **Forward vol:** The term structure of vols remains steep, so forward vols carry poorly. As a result, the path of the least resistance for forward vols is still lower.
- **Skews:** Short-dated risk reversals have cheapened and still signal a neutral view on duration. Long-dated receiver skew has cheapened from the recent highs.
- **Midcurves:** Midcurves on front end rates now trade rich to swaptions.
- **Curve vol:** The 10s30s gamma trades rich to recent realized vols and PCA.
- *Note: Futures trading involves substantial risk of loss.*

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See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures.

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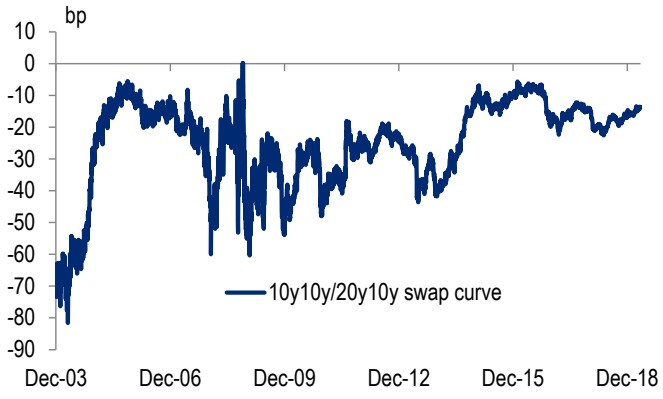
Weekly Focus: Trading long-dated convexity

Long-dated flatteners as a long convexity trade

In this note we explore ways to buy cheap volatility using very long-dated forward flatteners. To illustrate the idea, consider 10y10y/20y10y curve flatteners, structured by receiving 20y10y forward swaps vs paying 10y10y forward swaps, DV01-neutral. At that long horizons, expectations of monetary policy should flatten out, so the curve should be mostly determined by convexity adjustments, term premiums and technical factors. Because convexity is higher at longer maturities, the 20y10y forward should have a greater convexity per a unit of DV01 than 10y10y. As a result, the 10y10y/20y10 flattener must be positively convex.

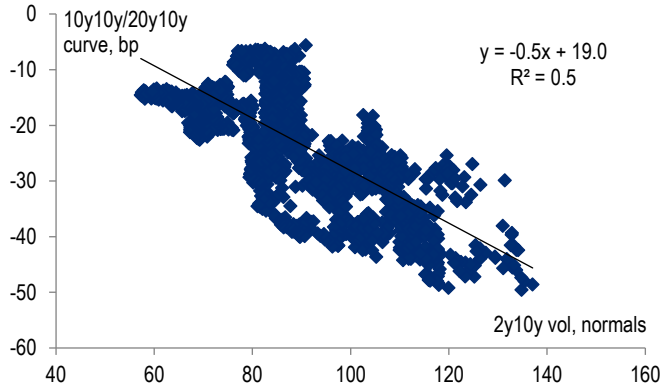
The flattener also rolls negatively given that the term structure of long-dated forwards is normally inverted (Figure 1). From that point of view, the risk profile of this trade is similar to that of long swaption straddles. In that analogy, the spread between 10y10y and 20y10y rates is akin to implied volatility, i.e. the vega exposure of the trade. Indeed, the 10y10y/20y10y curve is highly correlated empirically with implied rates vol (the higher the vol, the more inverted the curve is; Figure 2).

Figure 1. Very long-dated forward curve is normally inverted due to convexity



Source: Citi Research

Figure 2. The 10y10y/20y10y curve is highly (negatively) correlated with implied vols



Sample: 1/4/2010-5/3/2019. Source: Citi Research

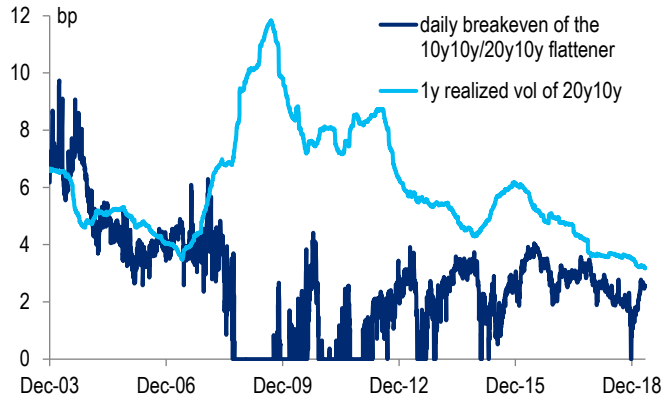
Long-dated curve flatteners can therefore be used to buy convexity, but how cheap is the embedded vol? One way to answer this question is to calculate implied daily rates breakeven, i.e. a parallel shift in rates that yields a convexity gain offsetting the negative daily roll. It turns out the daily breakeven calculated this way is typically small relative to realized vol. For example, the daily breakeven for the 10y10y/20y10y flattener has been consistently above the realized vol of corresponding forward rates (Figure 3; zero breakeven in the Figure 3 corresponds to episodes where the roll was positive).

This may be counterintuitive given the options analogy, since implied option vols typically exceed realized vols. Still, we think there are good reasons why this may be the case. Most importantly, curve flatteners do not offer positive convexity in the same clean way as options and therefore can be less attractive for convexity buyers. Indeed, unlike options, a curve flattener trade doesn't have a well-defined downside. Also, the flattener is difficult to use as a terminal trade. Convexity exposure declines while the volatility of the curve increases as the trade ages. As a result, the trade needs to be closed a few years after initiation, at which point an investor would realize any curve P&L (i.e. effectively the vega risk). In contrast, the vega risk is not relevant for the terminal P&L of options. Finally, we believe that life insurance flows may keep very long-dated curves structurally steep. This is because the duration of variable annuity portfolios is known to be concentrated in

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the 15-25y buckets. Structural receiving from insurance may therefore steepen the curve beyond 20y above the fair value.

Figure 3. Realized vol has typically exceed the implied daily breakeven



Note: The daily rates breakeven is a parallel shift in rates that yields a convexity gain offsetting the negative daily roll. When the roll is positive, the breakeven is zero.
Source: Citi Research

Figure 4. Systematically delta-hedged curve flatteners have been profitable, historically

curve	10y5y/ 15y15y	10y10y/ 15y15y	10y10y/ 20y10y	10y10y/ 20y15y	10y10y/ 25y10y	15y5y/ 20y10y	15y5y/ 20y15y	20y5y/ 25y10y
avg daily return, $K	0.67	1.15	2.28	3.20	5.40	2.92	3.79	4.09
daily vol of the P&L, $K	195.3	142.4	222.1	208.5	247.6	251.5	241.7	219.3
Sharpe	0.05	0.13	0.16	0.24	0.35	0.18	0.25	0.30

Note: For each pair of forward rates, we enter $100K DV01 of flatteners, rolled every year. We delta-hedge by adjusting the notional on the longer end of the trade at each 25bp move in the longer rate (at market close). Transaction costs assumptions can be found in the Appendix. Sample: 12/31/2013-5/7/2019. Past performance is no guarantee of future results. Source: Citi Research

Trading delta-hedged flatteners

The above observations suggest that systematic trading of long-dated forward flatteners can be a profitable strategy. This is indeed the case in our historical analysis. We analyze a strategy where we enter a DV01-neutral flattener rolled once a year. The full list of flatteners studied is shown in Figure 4. We start our analysis in 2004 because the quality of our swap curve for prior years is not satisfactory. Although we initiate a DV01-neural flattener, DV01 risks change as the market moves. We therefore delta-hedge the trade by adjusting the notional on the longer leg at each 25bp move in rates.

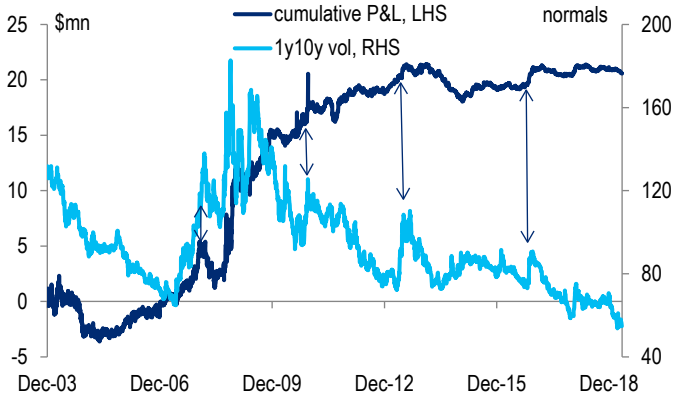
The 25bp threshold for delta-hedging was chosen as a trade-off between the accuracy of hedging and transaction costs. Delta hedging too frequently increases transaction costs, while hedging too infrequently increases the volatility of the P&L resulting in a smaller Sharpe (a similar trade-off exists in systematic gamma selling strategies; see for US Rates Vol Lab: The art of gamma hedging details). With our transaction costs assumptions, detailed in the Appendix, we have found that the Sharpe ratio of the strategy doesn't change significantly if the threshold is chosen in the 15-30bp range, but declines with a smaller or larger threshold. To simplify our historical analysis, we adjust the notional of the initial off-market swap, although delta-hedging with ATM swaps may be easier for the practical implementation of this strategy.

The strategy has been profitable for all long-dated curves in our sample (Figure 4). Systematic delta-hedging of 10y10y/25y10y flatteners delivered an especially attractive performance with the Sharpe of 0.35 (Figure 4). Although the Sharpe does not look very high, it is important to note that this strategy is structurally long volatility unlike some popular systematic carry trades, such as short gamma, monetizing term premium etc. Figure 5 shows that the trade tended to outperform during large spikes in volatility. The correlation of monthly P&L with the monthly changes in the 1y10y vol is about 26%. For the same reason, the distribution of daily P&L does not exhibit a significant negative tail (Figure 6). In fact, the skewness of daily P&L is about 0. In comparison, the popular strategy of systematically selling

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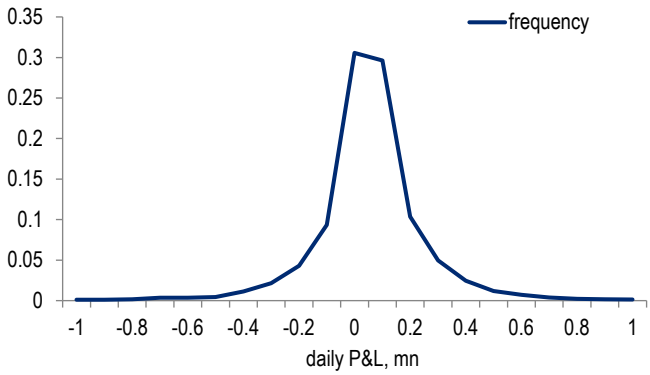
delta-hedged 1m10y straddles has a higher Sharpe of about 0.84, but also an extreme skewness of the P&L of about -3 (see US Rates Vol Lab: The art of gamma hedging for a detailed discussion of systematic short gamma).

Figure 5. The trade typically outperformed on vol spikes



Note: The chart shows the cumulative P&L of systematically delta-hedged 10y10y/25y10y flatteners vs 1y10y vol. Source: Citi Research

Figure 6. The P&L of the strategy is distributed roughly symmetrically



Note: The chart shows the distribution of the daily P&L of systematically delta-hedged 10y10y/25y10y flatteners. Sample: 12/31/2013-5/7/2019. Source: Citi Research

Despite the long vol bias, the strategy has outperformed even though implied vols have declined over our historical sample (Figure 5). The performance of the strategy has been largely flat since 2013 even though the 1y10y vol has declined by about 50 normals over the same time period (Figure 5). This reflects the cheapness of the embedded convexity relative to typical realized volatility.

The 15y5y/20y10y delta-hedged flatteners look attractive

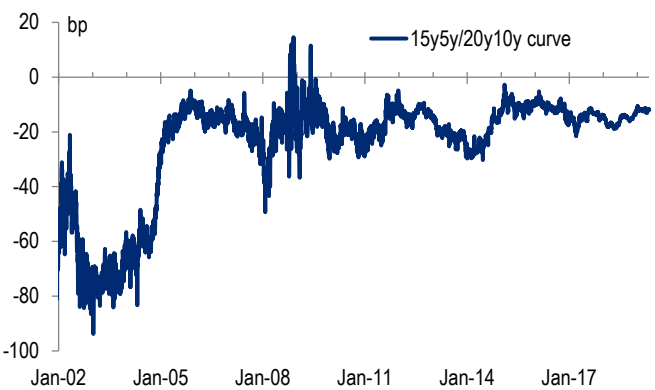
We believe the current environment is especially favorable for initiating delta-hedged curve flatteners. As discussed in US Rates Vol Lab: Vol cycle, steepeners, and SOFR, we believe rates vol is likely to see a cyclical turn higher in the next 6m-1y. At the same time, long-dated forward curves have steepened as a result of the recent decline in rates volatilities. In Figure 7 we evaluate a number of different flatteners by simple z-scores, carry, and implied daily breakeven. The last three trades look especially attractive on these metrics. We like the 15y5y/20y10y flattener in particular because it has a somewhat tighter market than the other two trades (15y5y/20y15y and 20y5y/25y10y).

Figure 7. The 15y5y/20y10y flatteners have an attractive entry point and implied convexity

curve	10y5y/ 15y15y	10y10y/ 15y15y	10y10y/ 20y10y	10y10y/ 20y15y	10y10y/ 25y10y	15y5y/ 20y10y	15y5y/ 20y15y	20y5y/ 25y10y
curve ,bp	-10.1	-8.8	-13.2	-16.7	-20.1	-11.3	-15.2	-9.1
1y ZS	2.50	2.15	1.87	2.41	2.38	1.16	1.62	1.95
3y ZS	0.19	0.39	0.53	0.50	0.55	**1.01**	**0.99**	**0.87**
1y carry, bp	-2.91	-1.64	-1.75	-1.88	-1.83	-0.31	-0.44	0.13
daily BE, bp	3.5	3.0	2.6	2.5	2.2	1.3	1.3	0.0
1y realized vol, bp	3.2	3.2	3.1	3.1	3.0	3.0	3.0	3.0
BE/realized vol	1.07	0.94	0.84	0.79	0.72	**0.42**	**0.45**	**0.00**

Note: Close of 5/8/2019. Source: Citi Research

Figure 8. The 15y5y/20y10y curve trades at the upper bound of its historical range



Source: Citi Research

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The 15y5y/20y10y curve is trading at the upper bound of its historical range (Figure 8). While the trade carries negatively by about -0.44bp per year, this negative carry is offset by just 1.3bp of daily volatility. In comparison, the daily volatility of the 20y10y rate has realized at 3bp and 2.8bp in the last year and the last three months, respectively. As discussed above, the trade needs to be hedged at every 15-30bp move in rates.

One risk to this trade is variable annuity hedging flows in a risk-off scenario. As discussed in North America Rates Focus: A new regime for 30y spreads, the duration of variable annuities extends when stocks underperform, so that insurance firms would typically need to receive in the ~20y sector in that scenario. Indeed, there exists a small negative correlation between stock returns and the 15y5y/20y10y curve of about -23% in the data (lower stocks leading to a steeper curve). However, we would also expect rates volatility to increase in that scenario, which should be positive for the gamma part of the trade P&L. In fact, there is little empirical correlation between the P&L of the trade and stock market performance (the correlation between monthly P&L of delta-hedged 15y5y/20y10y flatteners and monthly S&P returns has been close to 0 in our historical sample).

Appendix: Transaction costs assumptions

In Figure 9 we show our bid/offer assumptions for 1) initiating $100K DV01-neutral flatteners and 2) delta-hedging by adjusting the longer leg of the trade. While the market is quite wide for initiating flatteners of this size, delta hedging at each 25bp normally requires adjusting only a small size and therefore can be done at much tighter levels. Similarly, rolling the trade entails less risk for a dealer and can normally be executed at tighter levels. We assume the same bid/offer for the roll as for delta-hedging.

Figure 9. Transaction costs assumptions (one way)

	mid-to-offer and mid-to-bid, bp							
curve	10y5y/ 15y15y	10y10y/ 15y15y	10y10y/ 20y10y	10y10y/ 20y15y	10y10y/ 25y10y	15y5y/ 20y10y	15y5y/ 20y15y	20y5y/ 25y10y
initiation	0.75	0.75	0.75	1	1	0.75	1	1
delta-hedging and roll	0.3	0.3	0.3	0.4	0.4	0.3	0.4	0.4

Source: Citi Research

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At-the-money swaption vols

Levels and relative value on the surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- The vol surface has flattened across expiries and tails since we last published our Vol Lab. Upper-left vols have richened led by gamma, while longer tails have remained relatively unchanged (Figure 11).
- The 30y tail continues to look cheap on our macro-vol model and PCA of the surface (Figure 12 and Figure 13). For example, 3y30y is about two sigmas cheap to rates/curve and is 1.3 sigmas cheap on PCA (Figure 12).
- In contrast, upper-left vols look relatively rich on the surface. The 3m2y vol looks about 0.8 sigmas rich on our macro-vol framework (Figure 12).

Figure 10. Implied vols, normals (3m z-score)

exp\tail	1y	2y	3y	5y	7y	10y	30y
1m	33.8 (1.7)	52.6 (1.5)	55.9 (1.1)	59.2 (1.5)	58.5 (1.5)	56.1 (1.3)	48.4 (0.7)
3m	39.9 (1.4)	56.0 (1.8)	58.8 (1.6)	58.1 (1.4)	56.9 (1.2)	55.1 (1.0)	48.0 (0.2)
6m	48.9 (1.3)	58.7 (1.7)	60.5 (1.5)	60.7 (1.7)	58.9 (1.5)	57.2 (1.2)	49.9 (0.0)
1y	60.2 (1.5)	64.2 (1.5)	63.8 (1.7)	61.8 (1.0)	60.3 (0.9)	58.3 (0.4)	50.8 (-0.5)
2y	66.7 (1.3)	66.8 (0.6)	65.1 (0.4)	63.8 (0.2)	62.3 (0.0)	59.8 (-0.3)	52.4 (-0.7)
3y	67.6 (0.2)	67.9 (0.2)	66.5 (0.0)	64.3 (-0.3)	62.5 (-0.5)	60.4 (-0.6)	52.8 (-0.9)
5y	67.9 (-0.3)	67.6 (-0.4)	66.7 (-0.5)	64.6 (-0.7)	63.0 (-0.7)	61.3 (-0.7)	53.7 (-1.0)
10y	64.1 (-0.7)	63.5 (-0.8)	63.2 (-0.8)	62.3 (-0.8)	61.3 (-0.8)	59.2 (-0.8)	53.1 (-0.9)

Close of 5/8/19. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 11. 2w change in implied vols, normals

exp\tail	1y	2y	3y	5y	7y	10y	30y
1m	5.9	5.6	3.8	4.5	4.2	3.2	0.9
3m	5.5	6.8	6.0	3.5	3.1	2.6	1.2
6m	3.9	4.2	3.8	3.7	3.1	2.5	1.3
1y	3.9	3.5	3.7	2.8	2.8	2.1	1.0
2y	3.3	1.7	1.8	2.2	1.9	1.4	0.4
3y	2.1	1.5	1.3	1.0	0.6	0.8	0.0
5y	1.9	2.0	1.7	1.3	1.3	1.4	0.7
10y	1.5	1.4	1.2	0.9	0.9	0.9	0.8

Note: The table shows changes in ATM implied vols from 4/22/19 to 5/8/19. Source: Citi Research

Figure 12. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tail	1y	2y	3y	5y	7y	10y	30y
1m	2.3 (0.1)	11.1 (0.6)	7.8 (0.5)	4.3 (0.3)	4.1 (0.3)	3.2 (0.2)	-7.4 (-0.5)
3m	6.2 (0.3)	12.8 (0.8)	8.8 (0.6)	2.1 (0.2)	0.9 (0.1)	0.3 (0.0)	-9.3 (-0.8)
6m	11.5 (0.8)	11.3 (0.9)	7.4 (0.6)	2.3 (0.2)	0.6 (0.1)	-0.1 (0.0)	-9.1 (-1.0)
1y	12.9 (1.1)	8.8 (0.9)	5.0 (0.6)	-0.7 (-0.1)	-1.9 (-0.2)	-2.9 (-0.4)	-10.4 (-1.5)
2y	6.6 (0.8)	2.0 (0.3)	-0.9 (-0.1)	-3.0 (-0.4)	-3.8 (-0.6)	-5.1 (-0.8)	-11.2 (-1.8)
3y	0.4 (0.0)	-1.4 (-0.2)	-2.9 (-0.4)	-4.9 (-0.7)	-5.8 (-0.9)	-6.5 (-1.0)	-11.9 (-2.0)
5y	-3.1 (-0.4)	-3.9 (-0.6)	-4.7 (-0.7)	-6.2 (-1.0)	-6.9 (-1.2)	-7.3 (-1.3)	-11.4 (-2.0)
10y	-3.5 (-0.5)	-4.5 (-0.7)	-5.1 (-0.8)	-5.7 (-0.9)	-6.1 (-1.0)	-6.5 (-1.1)	-8.6 (-1.3)

Close of 5/8/19. We regress each ATM point on 3 principal components of the swap curve. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We used 1996-2008 spliced with 2017-2019 data to exclude the near-zero rate policy regime. Source: Citi Research

Figure 13. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tl	3m	1y	2y	3y	5y	7y	10y	30y
1m	-5.2 (-1.2)	0.5 (0.3)	1.3 (0.6)	0.8 (0.3)	2.7 (1.3)	2.4 (1.4)	1.1 (0.8)	-2.7 (-1.4)
3m	0.0 (0.0)	-1.2 (-0.7)	0.7 (0.5)	0.7 (0.4)	1.6 (1.3)	1.6 (1.7)	0.6 (0.9)	-2.4 (-2.0)
6m	2.9 (1.3)	0.3 (0.3)	0.3 (0.3)	0.1 (0.2)	1.8 (1.8)	1.9 (2.3)	1.1 (1.6)	-1.5 (-1.7)
1y	2.6 (1.8)	0.2 (0.2)	-0.3 (-0.3)	-0.1 (-0.2)	0.8 (1.1)	1.5 (2.3)	0.8 (1.1)	-1.2 (-1.7)
2y	-1.7 (-1.1)	-1.2 (-1.0)	-1.6 (-1.5)	-0.9 (-1.0)	-0.4 (-0.5)	0.3 (0.5)	0.2 (0.3)	-0.6 (-1.2)
3y	-2.1 (-1.5)	-1.8 (-1.6)	-1.6 (-1.5)	-1.1 (-1.4)	-0.4 (-0.8)	0.2 (0.4)	0.2 (0.5)	-0.4 (-1.3)
5y	-0.9 (-1.0)	-0.4 (-0.6)	-0.9 (-1.4)	-0.7 (-1.5)	0.0 (-0.1)	0.3 (0.9)	0.7 (1.6)	-0.2 (-0.6)
10y	-0.6 (-0.9)	-0.2 (-0.4)	-0.1 (-0.2)	0.1 (0.1)	0.5 (0.7)	0.7 (1.0)	0.9 (1.2)	0.8 (1.1)

Close of 5/8/19. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

Implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Intraday realized volatilities have slightly underperformed close-to-close vols in longer tails in the last month (Figure 14).
- Implieds generally trade rich to recent 1m realized vols (Figure 16 and Figure 17).
- While 30y gamma looks cheap on our macro vol model and the PCA of the vol surface, it still trades rich to the 1m realized vol (Figure 16).

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Figure 14. 1m intraday/close-to-close ratio of realized vols (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	1.13 (0.9)	1.03 (0.2)	1.02 (0.1)	0.97 (-0.3)	0.95 (-0.3)	0.95 (-0.3)	0.96 (-0.2)
3m	1.12 (0.7)	1.03 (0.2)	1.01 (0.1)	0.96 (-0.3)	0.95 (-0.3)	0.95 (-0.3)	0.96 (-0.2)
6m	1.07 (0.4)	1.01 (0.1)	0.99 (-0.1)	0.96 (-0.3)	0.95 (-0.4)	0.96 (-0.3)	0.96 (-0.2)
1y	1.02 (0.1)	1.01 (0.1)	0.97 (-0.2)	0.95 (-0.4)	0.94 (-0.4)	0.96 (-0.3)	0.95 (-0.3)

Note: Close of 5/8/19. Hourly vol between 8am and 3pm is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 15. 3m intraday/close-to-close ratio of realized vols (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	1.04 (0.7)	0.99 (-0.1)	1.01 (0.2)	1.02 (0.4)	1.00 (0.1)	1.00 (0.0)	1.00 (0.0)
3m	1.01 (0.2)	1.00 (-0.1)	1.01 (0.2)	1.02 (0.3)	1.00 (0.0)	1.00 (-0.1)	1.00 (0.0)
6m	0.98 (-0.3)	1.00 (-0.1)	1.01 (0.1)	1.02 (0.3)	1.00 (0.0)	1.00 (-0.1)	1.00 (0.0)
1y	1.01 (0.1)	1.02 (0.3)	1.02 (0.4)	1.02 (0.3)	1.00 (0.0)	1.00 (-0.1)	0.99 (-0.1)

Note: Close of 5/8/19. Hourly vol between 8am and 3pm is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 16. Intraday realized: implied/1m realized ratio (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	0.97 (-0.1)	1.10 (0.1)	1.04 (-0.2)	1.09 (0.0)	1.11 (0.2)	1.11 (0.2)	1.12 (0.5)
3m	0.84 (-0.4)	1.03 (-0.1)	1.03 (-0.3)	1.06 (-0.2)	1.09 (-0.1)	1.10 (0.0)	1.12 (0.2)
6m	0.90 (-0.3)	1.01 (-0.2)	1.02 (-0.4)	1.07 (-0.3)	1.10 (-0.1)	1.13 (0.0)	1.17 (0.2)
1y	0.94 (-0.3)	0.99 (-0.4)	1.03 (-0.4)	1.08 (-0.4)	1.12 (-0.2)	1.15 (-0.1)	1.19 (0.1)

Note: Close of 5/8/19. Hourly vol between 8am and 3pm is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 17. Intraday realized: implied/3m realized ratio (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	0.98 (0.1)	1.01 (-0.3)	0.95 (-0.6)	0.97 (-0.6)	0.98 (-0.5)	0.97 (-0.6)	0.94 (-0.6)
3m	0.84 (-0.3)	0.95 (-0.5)	0.93 (-0.8)	0.94 (-0.8)	0.96 (-0.8)	0.96 (-0.9)	0.94 (-0.9)
6m	0.87 (-0.3)	0.92 (-0.7)	0.93 (-0.9)	0.95 (-0.9)	0.97 (-0.9)	0.98 (-0.9)	0.98 (-0.9)
1y	0.83 (-0.8)	0.88 (-1.0)	0.92 (-1.0)	0.95 (-1.0)	0.98 (-1.0)	0.99 (-1.0)	1.00 (-0.9)

Note: Close of 5/8/19. Hourly vol between 8am and 3pm is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 18. Close-to-close realized: implied/1m realized ratio (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	1.11 (0.2)	1.13 (0.3)	1.06 (0.0)	1.05 (-0.1)	1.06 (0.0)	1.06 (0.0)	1.07 (-0.1)
3m	0.94 (-0.2)	1.06 (0.1)	1.04 (-0.1)	1.02 (-0.3)	1.03 (-0.2)	1.05 (-0.2)	1.08 (-0.2)
6m	0.97 (-0.1)	1.02 (-0.1)	1.02 (-0.3)	1.03 (-0.3)	1.05 (-0.3)	1.08 (-0.2)	1.11 (-0.2)
1y	0.95 (-0.1)	1.00 (-0.3)	1.01 (-0.4)	1.02 (-0.5)	1.06 (-0.4)	1.10 (-0.3)	1.13 (-0.3)

Note: Close of 5/8/19. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 19. Close-to-close realized: implied/3m realized ratio (1y z-score)

expr\tail	1y	2y	3y	5y	7y	10y	30y
1m	1.02 (0.1)	1.00 (0.0)	0.96 (-0.2)	0.99 (-0.1)	0.99 (-0.1)	0.96 (-0.4)	0.94 (-0.8)
3m	0.85 (-0.3)	0.95 (-0.2)	0.94 (-0.3)	0.96 (-0.4)	0.96 (-0.5)	0.95 (-0.7)	0.94 (-1.1)
6m	0.85 (-0.3)	0.92 (-0.3)	0.93 (-0.5)	0.97 (-0.5)	0.97 (-0.6)	0.98 (-0.7)	0.98 (-1.2)
1y	0.84 (-0.4)	0.90 (-0.5)	0.94 (-0.6)	0.96 (-0.7)	0.98 (-0.7)	0.99 (-0.9)	0.99 (-1.3)

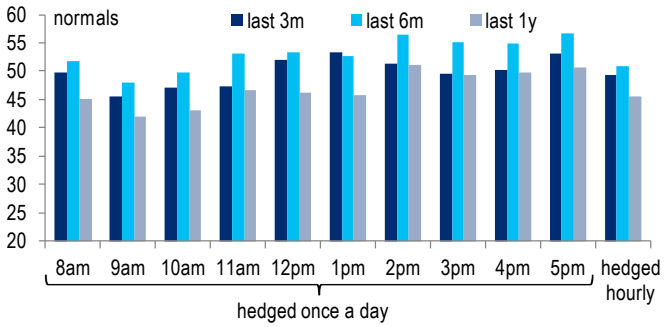
Note: Close of 5/8/19. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Day-to-day realized volatilities for a given time snapshot

We calculate day-to-day realized volatilities measured at a given hour within a day for the last 3m, 6m and 1y.

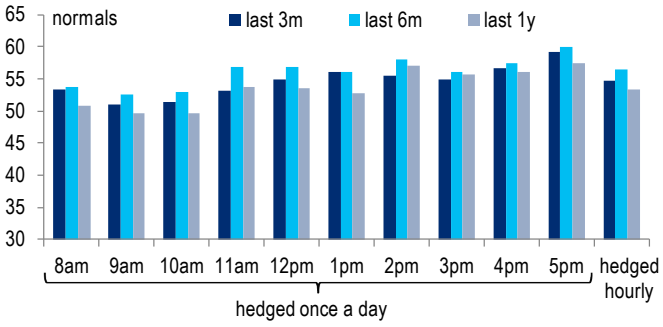
- Hedging around 9am would deliver one of the lowest realized volatilities over the past 3m, 6m and 1y. In contrast, hedging around 5pm delivered the highest realized vol.

Figure 20. Realized vol of the 2y rate



Note: Close of 5/8/19. Source: Citi Research

Figure 21. Realized vol of the 10y rate



Note: Close of 5/8/19. Source: Citi Research

Expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- The surface has flattened across expiries. The 1y10y/3y10y vol ratio is 2 sigmas high (Figure 22).
- The surface also has flattened across tails. The 2y5y/2y10y ratio is elevated by 1.8 sigmas (Figure 23).

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Figure 22. Short/long expiry ratio (3m z-score)

exp\tail	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.85 (-0.3)	0.94 (-0.6)	0.95 (-0.5)	1.02 (1.3)	1.03 (1.5)	1.02 (1.4)	1.01 (1.1)
3m/6m	0.82 (1.4)	0.95 (1.5)	0.97 (1.3)	0.96 (0.2)	0.97 (0.2)	0.96 (0.3)	0.96 (0.4)
6m/1y	0.81 (1.0)	0.91 (1.4)	0.95 (0.5)	0.98 (1.5)	0.98 (1.3)	0.98 (1.5)	0.98 (1.5)
1y/3y	0.89 (1.2)	0.94 (1.2)	0.96 (1.6)	0.96 (1.7)	0.97 (2.2)	0.96 (2.0)	0.96 (1.8)
3y/5y	1.00 (1.3)	1.00 (1.4)	1.00 (1.5)	1.00 (1.7)	0.99 (1.2)	0.99 (0.9)	0.98 (0.0)
5y/10y	1.06 (1.1)	1.06 (1.1)	1.05 (0.9)	1.04 (0.3)	1.03 (0.1)	1.04 (0.1)	1.01 (-0.9)

Note: Close of 5/8/19. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Figure 23. Short/long tail ratio (3m z-score)

exp\tail	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.64 (1.7)	0.60 (2.0)	0.89 (0.6)	0.94 (0.9)	1.05 (1.1)	1.22 (1.6)	1.16 (1.9)
3m	0.71 (0.9)	0.68 (1.2)	0.96 (1.6)	1.02 (1.5)	1.05 (1.2)	1.21 (1.6)	1.15 (1.8)
1y	0.94 (1.1)	0.94 (0.9)	1.04 (1.1)	1.10 (1.2)	1.06 (1.3)	1.21 (1.7)	1.15 (1.7)
2y	1.00 (1.8)	1.03 (1.6)	1.05 (0.9)	1.12 (1.3)	1.07 (1.8)	1.22 (1.7)	1.14 (1.5)
5y	1.00 (0.8)	1.02 (1.2)	1.05 (1.5)	1.10 (1.3)	1.05 (0.4)	1.20 (1.5)	1.14 (2.2)
10y	1.01 (1.2)	1.01 (1.0)	1.02 (-0.2)	1.07 (-0.2)	1.05 (0.0)	1.17 (0.0)	1.11 (0.0)

Note: Close of 5/8/19. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

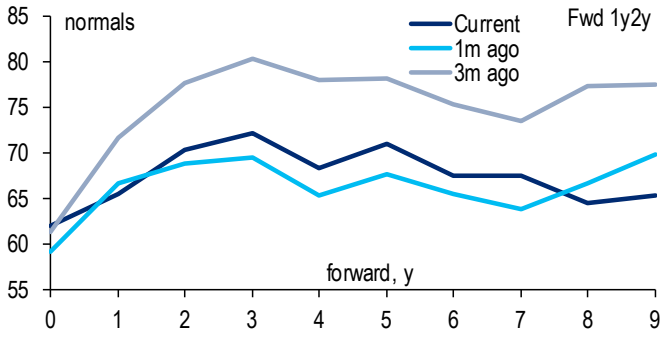
- We wrote in US Rates Vol Lab: Relative value in vega and curve vol that the term structure of long-dated forward vols is too steep. The surface has somewhat flattened since then as very long-dated forward vols have come off (Figure 27).
- The 4y point looks cheap on the term structure of forward 1y2y volatilities, which suggests 5y2y may be especially cheap on the spot surface (Figure 25). This is consistent with the PCA of the vol surface (Figure 13).

Figure 24. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y				1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	62.0	0.3	2.9	(1.0)	(0.2)	(9.9)	(0.4)	60.2	0.7	3.0	(5.3)	(1.3)	(25.5)	(1.1)	57.1	1.5	2.6	(6.9)	(1.7)	(32.6)	(1.5)	49.8	1.0	1.3	(9.3)	(2.0)	(35.2)	(1.8)
1	65.6	0.0	(1.2)	(7.4)	(1.1)	(6.3)	(0.2)	64.4	0.2	1.2	(7.6)	(1.6)	(21.3)	(0.9)	60.9	0.6	0.3	(8.3)	(2.0)	(28.8)	(1.3)	53.4	0.5	(0.1)	(9.7)	(2.2)	(31.6)	(1.6)
2	70.3	1.2	1.5	(7.2)	(1.1)	(1.5)	(0.1)	68.0	1.4	0.9	(8.1)	(1.6)	(17.7)	(0.8)	62.9	1.1	0.5	(8.5)	(2.0)	(26.8)	(1.2)	54.6	1.3	(0.6)	(8.9)	(2.2)	(30.3)	(1.5)
3	72.1	1.9	2.7	(9.2)	(1.3)	0.3	0.0	66.6	0.9	1.4	(8.9)	(1.6)	(19.1)	(0.8)	64.9	1.4	1.1	(8.4)	(1.8)	(24.9)	(1.1)	55.1	0.3	(0.6)	(8.1)	(2.3)	(29.8)	(1.5)
4	68.3	2.6	3.0	(10.2)	(1.6)	(3.6)	(0.1)	66.8	0.8	1.4	(9.9)	(1.7)	(18.9)	(0.8)	66.3	1.7	1.3	(8.1)	(1.8)	(23.4)	(1.1)	56.7	0.4	(0.5)	(6.6)	(2.2)	(28.2)	(1.4)
5	71.0	5.9	3.3	(5.3)	(1.1)	(0.9)	(0.0)	68.1	2.2	0.9	(8.1)	(1.6)	(17.6)	(0.8)	65.3	1.9	0.9	(7.2)	(1.8)	(24.4)	(1.1)	56.6	2.7	(0.0)	(4.6)	(1.7)	(28.4)	(1.4)
6	67.4	6.3	2.0	(11.5)	(1.4)	(4.4)	(0.2)	69.6	2.7	(0.1)	(5.8)	(1.5)	(16.1)	(0.7)	65.6	1.9	(0.1)	(7.1)	(1.9)	(24.1)	(1.1)	56.3	3.5	(0.0)	(3.3)	(1.4)	(28.7)	(1.4)
7	67.4	4.2	3.6	(1.5)	(0.4)	(4.5)	(0.2)	67.6	2.9	(1.8)	(7.6)	(2.0)	(18.1)	(0.8)	66.6	1.9	(1.9)	(7.1)	(2.1)	(23.1)	(1.1)	57.0	2.2	(1.0)	(2.3)	(1.5)	(27.9)	(1.4)
8	64.5	3.7	(2.2)	(12.3)	(1.9)	(7.4)	(0.3)	65.1	3.2	(2.6)	(7.6)	(2.0)	(20.6)	(0.9)	65.1	2.0	(2.9)	(6.7)	(2.2)	(24.6)	(1.1)	51.6	1.7	(1.4)	(1.5)	(0.9)	(33.4)	(1.7)
9	65.2	3.1	(4.5)	(10.6)	(2.7)	(6.6)	(0.3)	64.3	2.9	(1.2)	(4.4)	(1.8)	(21.4)	(0.9)	67.4	1.9	(1.8)	(2.7)	(1.7)	(22.3)	(1.0)	57.9	1.5	(0.7)	1.1	0.7	(27.0)	(1.3)

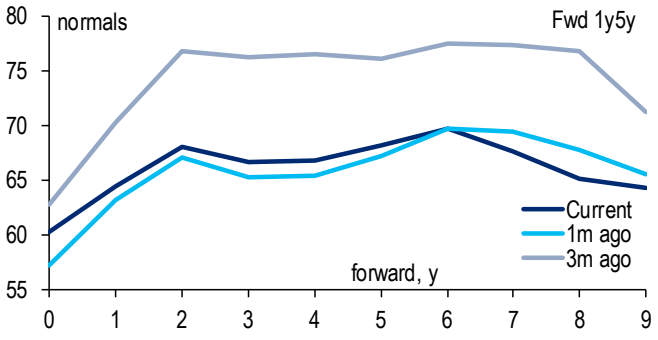
Note: Close of 5/8/19. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 25. Forward vol of 1y2y



Note: Close of 5/8/19. Source: Citi Research

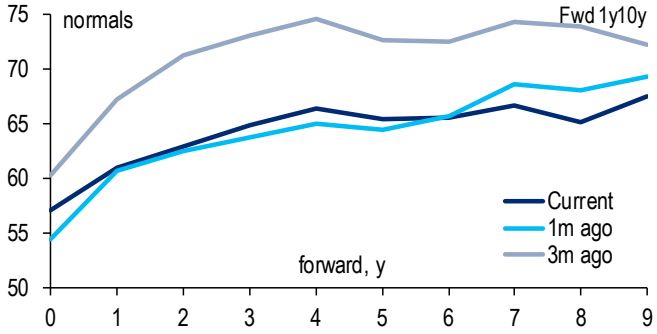
Figure 26. Forward vol of 1y5y



Note: Close of 5/8/19. Source: Citi Research

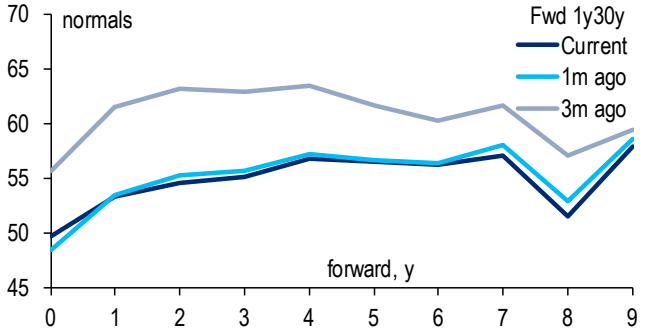
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Figure 27. Forward vol of 1y10y



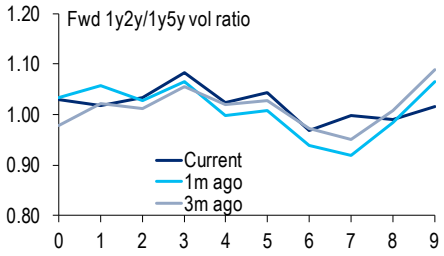
Note: Close of 5/8/19. Source: Citi Research

Figure 28. Forward vol of 1y30y



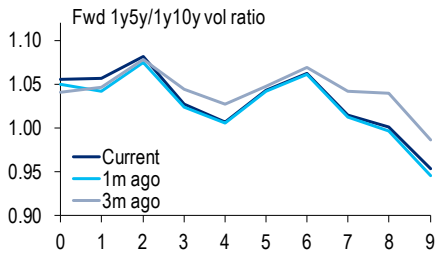
Note: Close of 5/8/19. Source: Citi Research

Figure 29. Fwd 1y2y/1y5y vol ratio



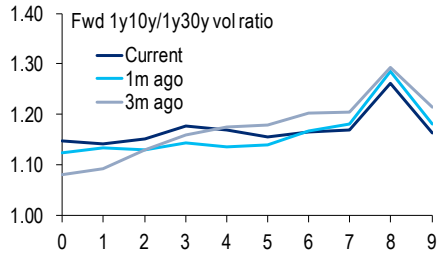
Note: Close of 5/8/19. Source: Citi Research

Figure 30. Fwd 1y5y/1y10y vol ratio



Note: Close of 5/8/19. Source: Citi Research

Figure 31. Fwd 1y10y/1y30y vol ratio



Note: Close of 5/8/19. Source: Citi Research

The efficient frontier of forward vol trades

We analyze all exactable forward vol trades, expressed using gamma-neutral calendar spreads and triangles, and report 20 trades on the efficient frontier based on ex-ante Sharpe (measured by the vol-adjusted roll) and value (measured by 1y-zcore).

- Carry in forward volatilities remains hard to get as the vol surface remains steep. Positive roll can only be found in ultra-long expiries (such as a 15y5y/15y30y/20y25y triangle; Figure 32). As a result, the path of the least resistance for forward vols may still be lower. See the US Rates Vol Lab: Relative value in vega and curve vol for more details.

Figure 32. Forward vol trades on the efficient frontier (Sharpe vs z-score)

short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS	short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS
15y5y/ 15y30y	20y25y	15y5y25y	94.8/ 100	50.3	50.5	51.2	0.32	0.28	1.7	0.9	2y5y25y	7y25y	2y5y25y	53.7/ 100	51.1	54.0	55.1	-0.15	-0.10	-2.2	-0.8
7y3y/ 7y5y	10y2y	7y3y2y	83.3/ 100	62.0	63.3	66.4	0.43	0.15	-2.0	-0.8	3y20y	10y5y	3y7y5y	12.5/ 100	54.0	62.1	66.0	-0.19	-0.12	-2.2	-0.8
3y30y	10y15y	3y7y15y	27.6/ 100	52.5	55.5	57.5	-0.04	-0.03	-2.1	-0.9	18m30y	7y20y	18m66m20y	29.6/ 100	50.8	54.5	56.0	-0.20	-0.14	-2.3	-0.9
2y5y/ 2y25y	7y20y	2y5y20y	50.8/ 100	49.4	54.5	56.5	-0.07	-0.04	-2.2	-0.8	2y25y	7y15y	2y5y15y	30.5/ 100	52.7	57.0	59.3	-0.22	-0.15	-2.2	-0.8
3y7y/ 3y10y	10y3y	3y7y3y	49.0/ 100	54.0	63.1	66.6	-0.08	-0.04	-2.1	-0.8	2y3y25y	5y25y	2y3y25y	62.1/ 100	51.3	53.9	55.6	-0.28	-0.18	-2.4	-1.1
2y5y20y	7y20y	2y5y20y	53.1/ 100	51.6	54.5	55.7	-0.11	-0.08	-2.2	-0.9	2y3y30y	5y30y	2y3y30y	63.1/ 100	51.2	53.4	54.7	-0.28	-0.19	-2.5	-1.2
2y30y	7y20y	2y5y20y	34.9/ 100	51.6	54.5	55.7	-0.11	-0.08	-2.3	-0.9	2y3y20y	5y20y	2y3y20y	61.8/ 100	52.2	54.7	56.4	-0.30	-0.19	-2.4	-1.1
2y5y/ 2y30y	7y25y	2y5y25y	51.1/ 100	48.5	54.0	56.0	-0.14	-0.09	-2.2	-0.8	3y25y	5y20y	3y2y20y	61.6/ 100	53.3	54.7	57.4	-0.33	-0.19	-2.5	-1.2
3y25y	10y10y	3y7y10y	21.0/ 100	53.3	59.0	62.0	-0.13	-0.09	-2.3	-0.9	3y30y	5y25y	3y2y25y	64.4/ 100	52.5	53.9	57.1	-0.37	-0.21	-2.5	-1.2
2y10y10y	7y20y	2y5y20y	105.8/ 100	50.9	54.5	55.9	-0.14	-0.10	-2.2	-0.9	3y30y	5y30y	3y2y30y	74.8/ 100	52.5	53.4	55.9	-0.36	-0.21	-2.6	-1.2

Note: Close of 5/8/19. The table reports 20 best forward vol trades on the efficient frontier based on ex-ante Sharpe ratio and 1y z-score. We analyze both calendar spreads (underlying forwards with matching mid-points) and triangles. The ex-ante Sharpe is calculated as vol-adjusted roll on the surface. Source: Citi Research

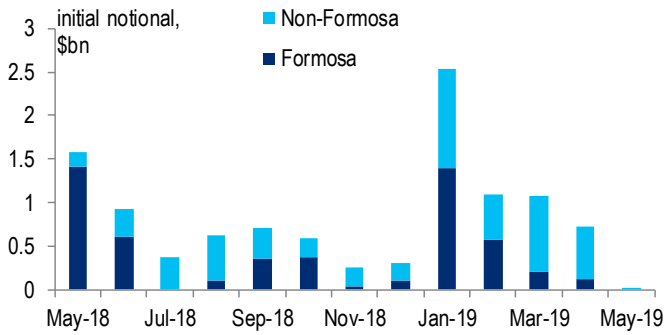
Prepared for Christopher Lee

Long-dated callable supply

We report supply and redemptions data for long-dated callable bonds.

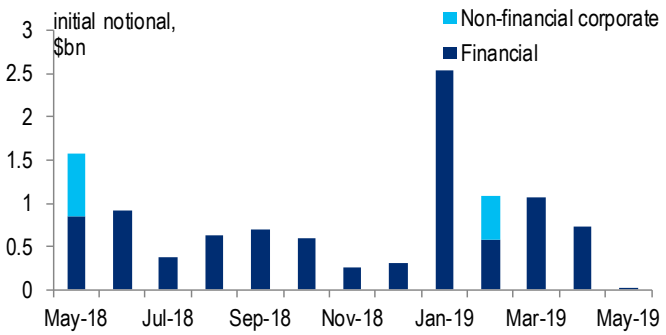
- Only about $730mn of long-dated callables was issued in April. We continue to expect only small supply going forward (roughly 25% of last year's supply).

Figure 33. Gross callable supply: Formosa vs Non-Formosa



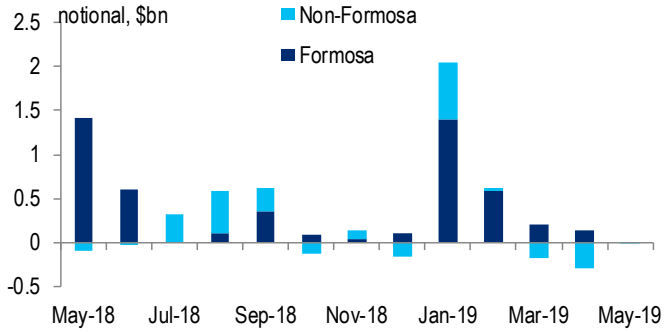
Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 34. Gross callable supply: Financial vs Non-Financial



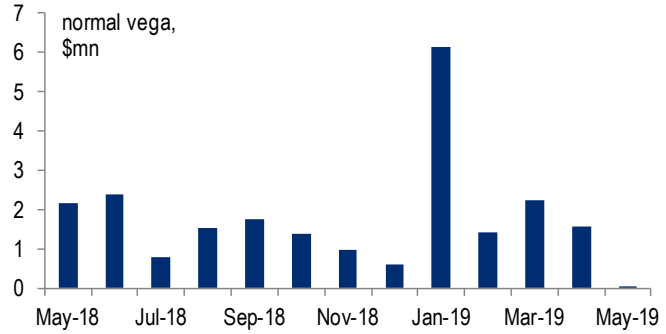
Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 35. Net callable notional supply



Source: Citi Research

Figure 36. Vega supply (normal parallel vega)



Note: We assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

Figure 37. Recent callable supply estimates

	gross issued, initial notional, $bn					redeemed, $bn					net issued, $bn					Vega, $mn
	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	
May-19 MTD	0.00	0.02	0.02	0.00	0.02	0.00	0.02	0.02	0.00	0.02	0.00	-0.01	-0.01	0.00	-0.01	0.0
Apr-19 (prev month)	0.13	0.60	0.73	0.00	0.73	0.00	0.90	0.90	0.00	0.90	0.13	-0.30	-0.17	0.00	-0.17	1.6
May-18 (prev year)	1.42	0.17	0.85	0.73	1.58	0.00	0.26	0.26	0.00	0.26	1.42	-0.10	0.59	0.73	1.32	2.2
2019 YTD	2.31	3.13	4.93	0.51	5.44	0.00	2.92	2.92	0.00	2.92	2.31	0.21	2.01	0.51	2.52	11.4
2018 YTD	17.02	3.07	17.96	2.13	20.09	1.04	2.35	3.03	0.37	3.39	15.98	0.71	14.93	1.77	16.69	43.9
2018 (all year)	18.61	5.28	21.76	2.13	23.89	1.33	3.73	4.70	0.37	5.06	17.27	1.55	17.06	1.77	18.83	53.3

Note: To compute vega we assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

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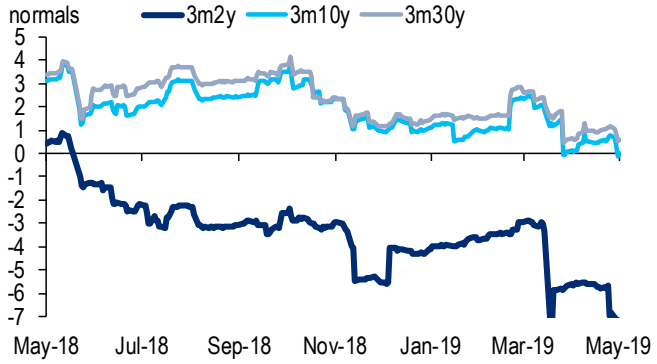
Swaption skew

Skew history and z-scores

We calculate z-scores of payer and receiver swaption skews across the surface.

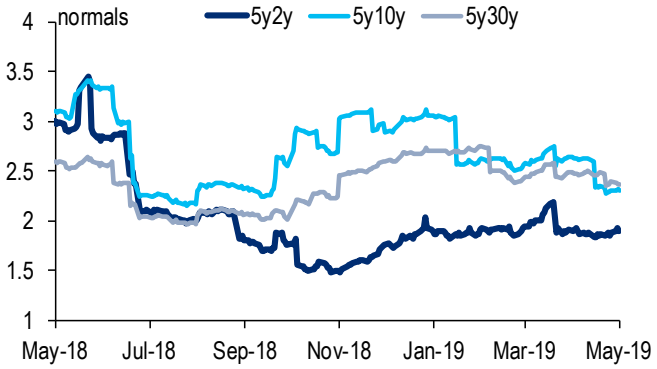
- Short-dated risk reversals have cheapened (Figure 38). Still, the 3m change in the 3m10y skew is about flat, implying a neutral signal for duration.
- Long-dated receiver skew has cheapened somewhat (Figure 39 and Figure 41).

Figure 38. 3m expiry payer skew



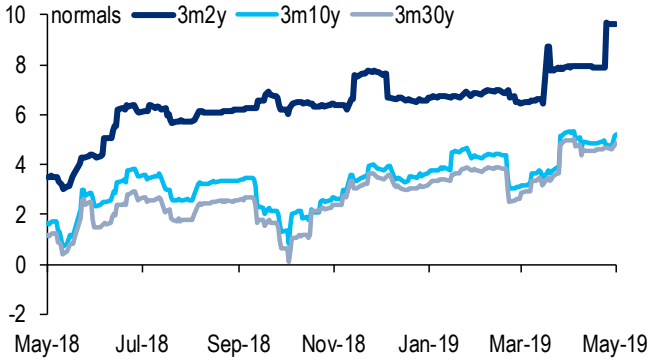
Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

Figure 39. 5y expiry payer skew



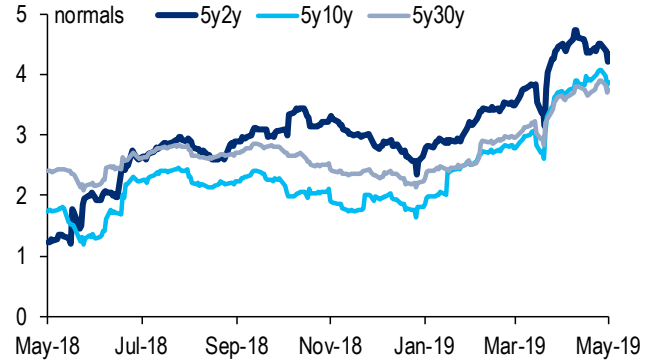
Note: 100bp-out payer skews for 5y expiry payers. Source: Citi Research

Figure 40. 3m expiry receiver skew



Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 41. 5y expiry receiver skew



Note: 100bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 42. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	-16.3 (-1.8)	-16.8 (-2.5)	-9.4 (-1.3)	-7.3 (-1.6)	-5.2 (-2.0)	-4.2 (-2.0)	-5.6 (-1.3)	-7.2 (-2.4)	-3.0 (-1.3)	-1.5 (-1.6)	0.0 (-2.0)	0.6 (-1.9)	10.7 (1.9)	9.6 (2.7)	6.4 (1.3)	5.8 (1.5)	5.2 (1.9)	4.8 (2.0)
6m	-26.2 (-2.1)	-23.8 (-2.2)	-13.3 (-1.7)	-9.9 (-1.7)	-6.6 (-1.7)	-5.1 (-2.0)	-7.4 (-1.8)	-8.9 (-2.3)	-3.4 (-1.7)	-0.8 (-1.7)	1.6 (-1.8)	2.8 (-1.9)	18.9 (2.0)	14.9 (2.2)	9.9 (1.6)	9.0 (1.6)	8.1 (1.7)	7.9 (1.9)
1y	-19.2 (-1.8)	-16.5 (-1.9)	-11.1 (-2.1)	-7.9 (-1.9)	-5.0 (-1.8)	-4.0 (-1.9)	-8.0 (-2.1)	-7.3 (-2.0)	-3.8 (-2.2)	-1.7 (-2.0)	0.1 (-1.8)	1.2 (-1.7)	11.2 (1.4)	9.1 (1.6)	7.4 (2.0)	6.2 (1.9)	5.2 (1.8)	5.2 (2.0)
2y	-8.8 (-1.5)	-7.9 (-1.7)	-5.1 (-2.3)	-4.7 (-2.3)	-3.7 (-2.3)	-2.8 (-2.9)	-3.4 (-1.4)	-3.0 (-1.6)	-1.6 (-2.2)	-1.2 (-2.3)	-0.4 (-2.4)	0.0 (-3.3)	5.4 (1.6)	4.9 (1.8)	3.5 (2.3)	3.5 (2.3)	3.3 (2.3)	2.9 (2.7)
3y	-9.7 (-1.4)	-7.8 (-1.5)	-5.8 (-2.3)	-4.6 (-2.4)	-4.1 (-2.3)	-2.9 (-3.1)	-1.8 (-1.0)	-0.6 (-1.1)	0.8 (-2.0)	1.2 (-2.1)	1.9 (-2.1)	2.4 (-3.3)	7.9 (1.7)	7.2 (1.8)	6.6 (2.4)	5.8 (2.4)	5.9 (2.3)	5.4 (2.8)
5y	-3.5 (-1.0)	-2.4 (-1.3)	-1.9 (-1.4)	-1.3 (-1.8)	-1.6 (-2.2)	-1.4 (-2.0)	1.4 (-0.2)	1.9 (-0.3)	2.1 (-0.6)	1.7 (-1.0)	2.3 (-1.2)	2.4 (-0.1)	4.9 (1.5)	4.3 (1.8)	3.9 (1.9)	3.0 (2.1)	3.9 (2.4)	3.7 (2.5)
10y	1.1 (-1.1)	0.6 (-1.3)	0.1 (0.5)	-0.2 (-1.0)	-0.9 (-2.0)	-0.4 (0.3)	2.4 (-1.1)	2.1 (-1.4)	2.2 (0.5)	1.9 (-0.9)	1.6 (-1.8)	2.2 (0.2)	1.3 (1.1)	1.6 (1.2)	2.0 (-0.2)	2.1 (0.7)	2.5 (1.5)	2.6 (-0.5)

Note: Close of 5/8/19. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. The following OTM conventions are used: 25bp out for 3m expiries, 50bp out for 6m-2y expiries, 100bp out for 3y+ expiries. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

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Implied vs realized skew

We compare implied swaption skews with realized skews. We calculate realized skews as the model value of OTM options from the empirical SABR model calibrated to the recent (3m) realized dynamics of ATM volatilities and rates.

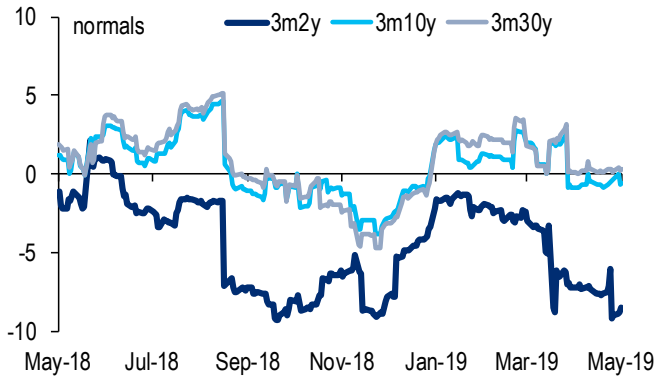
- Risk reversals in long expiries continue to trade rich to realized skew. For example, the 5y30y risk reversals trade more than 2.7 sigmas rich (Figure 43 and Figure 45).

Figure 43. Implied vs realized skews (spread to the model for risk reversals, price/model ratio for payers and receivers)

	Risk Reversals (1Y z-score)					OTM payers (1Y z-score)					OTM receivers (1Y z-score)				
expr\tail	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y
3m	-4.2 (0.0)	-8.5 (-1.4)	-3.8 (-0.5)	-0.6 (-0.6)	0.3 (-0.2)	0.98 (-0.2)	0.93 (-1.5)	0.99 (-0.7)	1.04 (-0.4)	1.06 (0.2)	1.08 (-0.5)	1.09 (0.7)	1.06 (0.3)	1.05 (0.6)	1.05 (0.5)
6m	-8.0 (-0.5)	-10.7 (-1.2)	-3.6 (-0.4)	1.6 (0.3)	3.2 (0.9)	1.01 (-0.4)	0.94 (-1.3)	1.02 (-0.6)	1.10 (1.0)	1.14 (1.6)	1.15 (0.2)	1.13 (0.7)	1.07 (0.1)	1.06 (-0.1)	1.06 (-0.4)
1y	-4.4 (0.0)	-5.2 (-0.5)	-3.1 (-0.5)	1.2 (0.4)	2.5 (1.4)	0.98 (-0.4)	0.97 (-1.0)	1.00 (-0.7)	1.05 (1.6)	1.09 (2.4)	1.05 (-0.5)	1.05 (0.0)	1.05 (0.4)	1.03 (0.0)	1.03 (-0.5)
2y	-0.7 (1.5)	-0.8 (1.0)	-0.1 (0.4)	0.4 (0.1)	1.4 (1.1)	1.01 (1.6)	1.01 (1.1)	1.01 (0.7)	1.03 (0.9)	1.04 (1.8)	1.02 (-1.4)	1.02 (-0.9)	1.01 (-0.2)	1.02 (0.3)	1.01 (-0.7)
3y	-1.7 (0.9)	0.1 (0.9)	1.1 (0.8)	2.3 (1.1)	3.0 (1.5)	1.03 (1.3)	1.05 (1.6)	1.07 (1.8)	1.08 (2.2)	1.09 (2.1)	1.05 (-0.3)	1.04 (-0.4)	1.04 (-0.2)	1.03 (-0.4)	1.03 (-1.2)
5y	2.4 (2.3)	3.1 (2.4)	4.1 (2.4)	4.5 (2.5)	4.6 (2.7)	1.06 (3.1)	1.07 (3.1)	1.08 (2.8)	1.09 (2.7)	1.10 (2.9)	1.03 (-1.2)	1.02 (-1.3)	1.01 (-2.1)	1.01 (-2.3)	1.00 (-2.8)
10y	6.6 (2.5)	5.7 (3.2)	4.9 (3.0)	3.7 (2.9)	4.6 (2.8)	1.08 (2.7)	1.07 (3.2)	1.07 (2.9)	1.06 (2.6)	1.09 (2.7)	0.97 (-2.8)	0.98 (-3.5)	0.99 (-3.4)	1.00 (-3.3)	1.00 (-3.1)

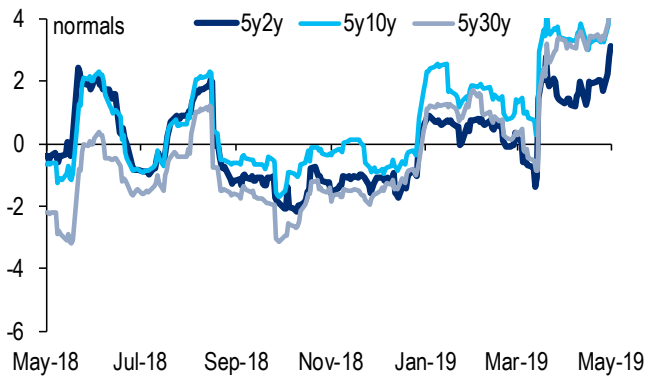
Note: Close of 5/8/19. We calibrate an empirical SABR model to the recent observed dynamics of ATM vols and rates. We use a 2y rolling window to estimate a backbone parameter, and a 3m window to estimate correlation and vol-of-vol parameters. The empirical model matches ATM implied vols exactly, so it's different from the market only in skew parameters. We calculate "realized skew" as the theoretical values of OTM payers, receivers and risk reversals from this empirical model. The table shows market-"realized" spreads of risk reversals, and market/"realized" ratios of OTM payers and receivers together with 1y z-scores of these spreads and ratios in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 44. 3m expiry risk reversals/model spread



Note: The table shows model-"realized" spreads for 25bp-out risk reversals in 3m expiries using the same methodology as in Figure 43. Source: Citi Research.

Figure 45. 5y expiry risk reversals/model spread



Note: The table shows model-"realized" spreads for 100bp-out risk reversals in 5y expiries using the same methodology as in Figure 43. Source: Citi Research.

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Board/swaption relative value

We evaluate the relative pricing of Board and swaption volatilities.

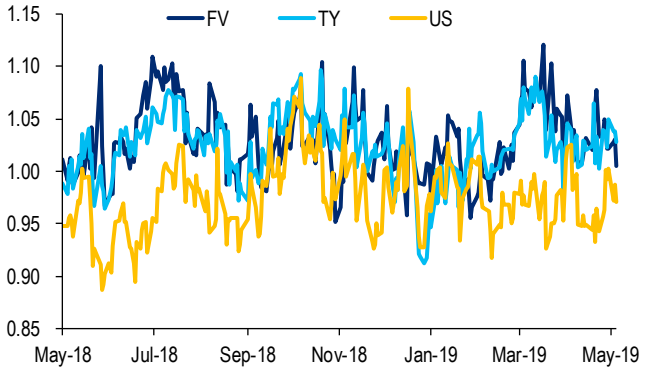
- TYN9 vol trades only 3% rich to swaptions, which looks slightly low when compared to the ratios of recent realized vols (Figure 46 and Figure 47).
- *Note: Futures trading involves substantial risk of loss.*

Figure 46. Board vols vs swaptions

option	FVM9	FVN9	TYM9	TYN9	USM9	USN9
Board vol	62.2	60.7	60.1	60.1	49.5	49.6
1w chg	8.5	2.7	6.0	4.9	5.5	2.5
Board/ 1m RV	1.12	1.03	1.02	0.97	1.03	1.03
Board/ 3m RV	1.04	1.00	0.99	0.96	0.93	0.94
CTD-matched swpn vol	56.2	60.4	55.4	58.5	48.6	51.1
1w chg	4.6	4.9	4.5	5.3	5.2	4.1
swpn/ 1m RV	1.01	1.03	0.99	1.02	1.02	1.05
swpn/ 3m RV	0.95	0.97	0.94	0.96	0.92	0.95
Board/swpn	1.11	1.01	1.08	1.03	1.02	0.97
1w ago	1.04	1.05	1.06	1.04	1.01	1.00
ftrs 1m RV/swap 1m RV	1.00	1.00	1.05	1.08	1.00	0.99
ftrs 3m RV/swap 3m RV	1.01	0.97	1.02	1.04	1.01	0.99

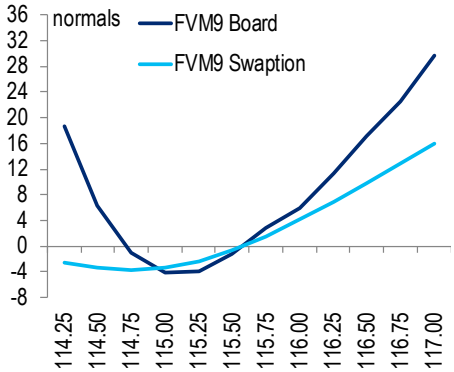
Note: Close of 5/8/19. The table shows implied ATM (interpolated) Board vols vs CTD-matched swaption vols with corresponding realized volatilities. Source: Citi Research

Figure 47. Board/swaption vol ratio



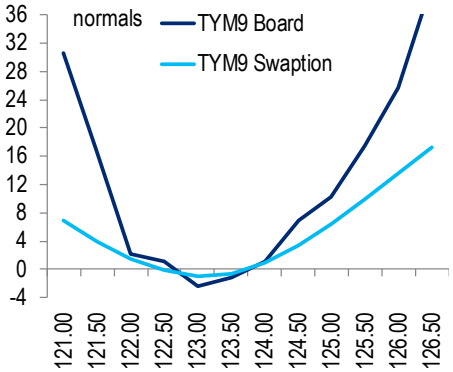
Note: The chart shows the ratio of ATM Board vol to CTD-matched swaption vols. The second available expiry (rolled on the expiration of front options) is used for the chart. Source: Citi Research

Figure 48. FV Board vs swaption skew



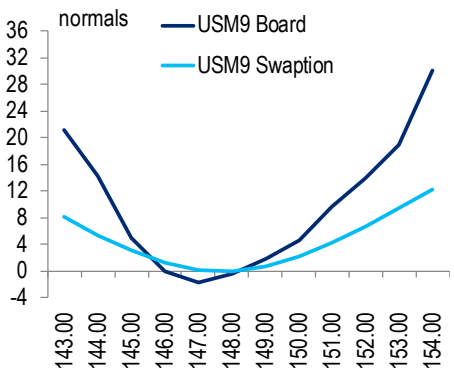
Note: Close of 5/8/19. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

Figure 49. TY Board vs swaption skew



Note: Close of 5/8/19. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

Figure 50. US Board vs swaption skew



Note: Close of 5/8/19. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

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Figure 51. Conditional spread trades

							puts/payers								calls/receivers							
optn	expiry	futures price	fut DV01	swap DV01	swpn ntl, $mn	invc sprd, bp	put strike	strike yield, %	costless payer strike,	strike spread, bp	bp pick-up	put vol	payer vol	vol ratio	call strike	strike yield, %	costless receiver strike, %	strike spread, bp	bp pick-up	call vol	rcvr vol	vol ratio
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	115.50	2.28	2.30	1.6	1.1	60.9	56.0	1.09	115.75	2.23	2.28	4.7	-1.9	65.0	57.0	1.14
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	115.25	2.34	2.35	1.5	1.3	58.2	54.3	1.07	116.00	2.17	2.23	5.5	-2.7	68.2	59.0	1.16
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	115.00	2.40	2.40	0.8	2.0	58.1	53.2	1.09	116.25	2.12	2.19	7.1	-4.3	73.5	60.9	1.21
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	114.75	2.45	2.44	-1.0	3.8	61.2	52.7	1.16	116.50	2.06	2.16	9.1	-6.4	79.4	62.6	1.27
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	114.50	2.51	2.46	-4.6	7.3	68.5	52.5	1.30	116.75	2.01	2.12	11.2	-8.5	84.6	64.4	1.31
FVM9	05/24/19	115-18+	4.53	3.95	114.5	2.8	114.25	2.56	2.46	-10.2	12.9	80.9	52.5	1.54	117.00	1.96	2.10	14.3	-11.6	91.8	65.7	1.40
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	115.50	2.28	2.32	4.1	0.4	60.1	59.3	1.01	115.75	2.22	2.27	4.7	-0.2	60.9	60.8	1.00
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	115.25	2.33	2.38	4.5	0.0	57.4	57.8	0.99	116.00	2.17	2.22	5.2	-0.7	64.0	62.6	1.02
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	115.00	2.39	2.43	4.3	0.2	56.9	57.0	1.00	116.25	2.11	2.17	5.4	-0.9	66.3	64.6	1.03
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	114.75	2.44	2.48	4.1	0.5	56.9	56.5	1.01	116.50	2.06	2.12	6.3	-1.7	69.7	66.6	1.05
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	114.50	2.50	2.52	2.4	2.1	59.5	56.4	1.05	116.75	2.00	2.07	6.6	-2.1	72.2	68.9	1.05
FVN9	06/21/19	115-21.75	4.52	3.93	115.2	4.5	114.25	2.56	2.56	0.6	3.9	62.0	56.5	1.10	117.00	1.95	2.03	8.0	-3.4	76.3	70.7	1.08
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	123.50	2.40	2.38	-2.6	1.0	58.9	55.0	1.07	124.00	2.33	2.33	-0.2	-1.3	61.3	55.9	1.10
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	123.00	2.47	2.44	-2.8	1.3	57.7	54.4	1.06	124.50	2.27	2.28	1.4	-2.9	67.0	57.6	1.16
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	122.50	2.54	2.50	-4.4	2.8	61.2	54.8	1.12	125.00	2.20	2.23	2.5	-4.0	70.3	59.9	1.17
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	122.00	2.61	2.56	-5.0	3.5	62.2	56.0	1.11	125.50	2.13	2.19	5.2	-6.8	77.6	61.8	1.26
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	121.50	2.68	2.56	-11.9	10.4	76.6	56.0	1.37	126.00	2.07	2.15	8.8	-10.3	85.8	63.3	1.36
TYM9	05/24/19	123-24	7.37	6.03	122.2	-1.5	121.00	2.75	2.56	-18.8	17.3	90.6	56.0	1.62	126.50	2.00	2.15	15.4	-17.0	100.5	63.3	1.59
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	123.50	2.43	2.41	-1.6	0.6	58.9	57.6	1.02	124.00	2.36	2.36	-0.4	-0.6	60.2	58.4	1.03
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	123.00	2.49	2.48	-1.3	0.3	57.6	57.3	1.00	124.50	2.29	2.29	-0.3	-0.7	61.4	60.0	1.02
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	122.50	2.56	2.54	-2.0	1.0	59.3	57.7	1.03	125.00	2.23	2.23	0.7	-1.7	65.2	61.8	1.06
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	122.00	2.63	2.61	-2.1	1.1	60.2	58.8	1.02	125.50	2.16	2.17	1.4	-2.4	68.2	64.0	1.07
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	121.50	2.70	2.66	-3.2	2.2	63.0	60.1	1.05	126.00	2.10	2.12	2.2	-3.2	71.5	66.3	1.08
TYN9	06/21/19	123-31+	7.52	6.07	123.9	-1.0	121.00	2.76	2.71	-5.5	4.5	67.0	61.4	1.09	126.50	2.03	2.08	4.8	-5.8	77.1	68.0	1.13
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	148.00	2.64	2.61	-3.0	0.3	49.0	48.6	1.01	149.00	2.58	2.56	-2.0	-0.7	51.3	49.1	1.04
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	147.00	2.69	2.66	-2.7	0.0	47.8	48.8	0.98	150.00	2.53	2.51	-1.3	-1.5	54.1	50.3	1.08
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	146.00	2.75	2.72	-3.0	0.3	49.5	49.8	0.99	151.00	2.47	2.48	0.3	-3.1	59.1	51.6	1.14
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	145.00	2.80	2.76	-4.6	1.9	54.5	51.0	1.07	152.00	2.42	2.44	1.9	-4.7	63.4	53.2	1.19
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	144.00	2.86	2.77	-8.7	6.0	63.7	51.6	1.24	153.00	2.37	2.41	4.0	-6.8	68.4	54.6	1.25
USM9	05/24/19	148-7	18.22	13.75	132.5	-2.7	143.00	2.92	2.80	-12.2	9.4	70.7	52.5	1.35	154.00	2.31	2.41	9.3	-12.0	79.6	54.6	1.46
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	147.00	2.66	2.63	-2.6	-0.5	49.1	50.9	0.97	148.00	2.60	2.57	-3.4	0.3	49.9	51.4	0.97
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	146.00	2.72	2.69	-2.3	-0.8	48.9	51.1	0.96	149.00	2.55	2.52	-2.9	-0.1	52.1	52.4	1.00
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	145.00	2.77	2.76	-1.7	-1.4	49.1	52.1	0.94	150.00	2.49	2.47	-2.6	-0.5	54.3	53.8	1.01
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	144.00	2.83	2.82	-0.7	-2.4	49.5	53.9	0.92	151.00	2.44	2.42	-2.1	-0.9	56.7	55.5	1.02
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	143.00	2.89	2.87	-2.0	-1.1	53.2	55.3	0.96	152.00	2.38	2.37	-1.9	-1.1	58.8	57.5	1.02
USN9	06/21/19	147-18	17.93	13.50	132.8	-3.1	142.00	2.95	2.90	-4.5	1.5	58.2	56.5	1.03	153.00	2.33	2.32	-1.0	-2.1	62.1	59.3	1.05

Note: Close of 5/8/19. Swaption notional is for 1000 Board contracts. The costless swaption strike makes the DV01-neutral trade costless. The BP pick-up is the spread between the costless strike spread and invoice spread. Futures trading involves substantial risk of loss. Source: Citi Research

Midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Short-dated midcurves on front end rates now trade rich to swaptions. This can be seen from the decline of implied correlations and direct regressions. For example, the 3m2y1y vol looks about 1.7 normals rich to swaptions (Figure 55).
- In contrast, midcurves on 10y10y look rich to swaptions (Figure 55).

Figure 52. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	73.6 (0.8)	73.0 (0.8)	66.7 (0.7)	73.1 (0.8)	69.6 (0.8)	64.7 (0.8)	51.7 (0.0)	44.7 (-0.3)
3m	69.4 (1.0)	67.2 (0.8)	63.5 (1.0)	68.1 (0.9)	64.8 (0.9)	61.5 (1.0)	52.5 (-0.2)	45.4 (-0.7)
6m	68.6 (1.1)	65.9 (0.8)	61.5 (0.5)	66.8 (1.0)	63.3 (0.7)	59.9 (0.7)	55.2 (-0.3)	47.2 (-0.9)
1y	68.9 (0.4)	64.6 (0.3)	62.2 (0.3)	66.2 (0.4)	63.1 (0.6)	61.8 (0.4)	56.7 (-0.6)	48.9 (-1.0)
2y	65.0 (-0.2)	61.7 (-0.4)	59.8 (-0.5)	63.1 (-0.3)	62.3 (0.0)	59.1 (-0.5)	60.3 (-0.9)	50.9 (-1.0)

Close of 5/8/19. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 53. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	87.2 (-0.3)	94.9 (-1.3)	99.0 (-0.5)	52.9 (-1.2)	89.7 (-1.0)	97.9 (-0.1)	98.3 (1.9)	97.6 (0.3)
3m	98.5 (0.7)	98.2 (-0.9)	99.3 (-0.5)	88.5 (-0.4)	95.9 (-0.9)	98.2 (-0.4)	98.0 (1.6)	98.3 (0.9)
6m	97.0 (-0.2)	99.0 (-0.6)	99.9 (0.4)	92.7 (-0.5)	98.6 (-0.2)	99.7 (0.3)	97.9 (1.6)	97.9 (0.5)
1y	97.5 (-0.5)	99.2 (0.1)	99.5 (-0.4)	94.6 (-0.3)	97.7 (-0.4)	98.4 (-0.7)	97.9 (1.9)	97.8 (0.2)
2y	100.2 (0.0)	100.1 (1.1)	99.9 (0.2)	101.0 (0.5)	98.8 (0.2)	100.3 (0.3)	96.5 (-0.3)	97.6 (0.1)

Close of 5/8/19. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 54. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	0.90 (-0.6)	0.95 (-1.4)	0.99 (-0.6)	0.56 (-1.4)	0.91 (-1.0)	0.99 (-0.4)	1.00 (0.7)	0.99 (-1.0)
3m	1.00 (0.5)	0.99 (-0.8)	1.00 (-1.0)	0.92 (-0.2)	0.97 (-1.0)	0.99 (-0.8)	0.99 (0.3)	0.99 (-0.3)
6m	0.98 (-0.1)	0.99 (-0.7)	1.00 (0.7)	0.95 (-0.5)	1.00 (-0.1)	1.01 (0.3)	0.99 (0.4)	0.99 (-0.9)
1y	0.98 (0.4)	1.00 (0.3)	1.00 (-0.2)	0.96 (0.6)	0.99 (-0.3)	0.99 (-0.7)	0.99 (0.5)	0.99 (-1.2)
2y	1.01 (-1.9)	1.00 (1.9)	1.00 (0.4)	1.02 (-0.1)	0.99 (-0.4)	1.01 (0.8)	0.97 (-0.7)	0.98 (-1.7)

Close of 5/8/19. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 55. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	2.8 (0.7)	4.2 (1.2)	0.0 (0.0)	4.6 (1.3)	2.6 (0.8)	-1.7 (-0.5)	-1.4 (-1.2)	1.4 (1.1)
3m	-0.5 (-0.2)	1.7 (0.7)	-0.5 (-0.2)	0.9 (0.4)	1.2 (0.5)	-1.0 (-0.4)	0.0 (0.0)	1.1 (1.1)
6m	-0.4 (-0.2)	1.0 (0.5)	-3.6 (-1.4)	0.5 (0.3)	-1.1 (-0.5)	-2.6 (-1.0)	-0.1 (-0.1)	1.9 (1.8)
1y	-3.0 (-1.7)	-2.5 (-1.1)	-1.5 (-0.6)	-2.4 (-1.3)	-1.3 (-0.5)	-0.1 (-0.1)	-0.3 (-0.6)	2.4 (2.3)
2y	-2.0 (-0.8)	-2.1 (-1.0)	-0.3 (-0.1)	-1.6 (-0.8)	0.2 (0.1)	-1.1 (-0.4)	0.6 (1.1)	1.6 (1.6)

Close of 5/8/19. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

Curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- Curve gamma trades at a higher premium to recent realized vols than swaptions (Figure 59). The 10s30s vol looks especially rich on PCA and relative to recent realized vols (Figure 57 and Figure 59).

Figure 56. Implied vols, 15ormal (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	18.4 (0.5)	25.3 (0.5)	28.4 (0.5)	13.8 (0.9)	25.0 (1.3)	15.6 (1.0)
3m	18.0 (0.1)	25.7 (0.2)	29.5 (0.4)	13.7 (0.7)	24.8 (1.4)	15.6 (1.1)
6m	18.7 (0.4)	27.8 (0.6)	32.1 (0.8)	14.2 (0.7)	24.8 (1.0)	16.3 (1.3)
1y	20.2 (0.5)	29.8 (0.2)	35.8 (0.4)	14.6 (0.0)	27.0 (0.5)	16.7 (1.5)
2y	21.2 (-0.3)	33.2 (-0.7)	38.0 (-0.5)	16.2 (-0.4)	28.7 (0.9)	15.8 (0.0)
3y	21.3 (-0.4)	34.0 (-0.6)	38.9 (-0.6)	16.0 (-0.5)	29.2 (0.5)	16.0 (-0.4)
5y	19.2 (-0.6)	32.9 (-0.7)	38.5 (-0.6)	14.7 (-0.7)	27.1 (-0.6)	16.4 (-0.4)

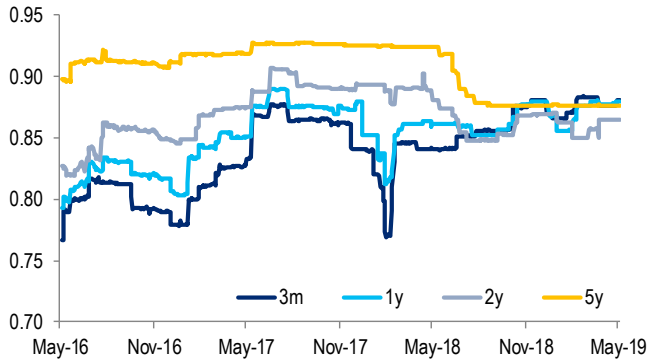
Close of 5/8/19. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 57. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	0.0 (0.0)	-1.9 (-0.9)	-2.0 (-0.8)	0.1 (0.0)	2.6 (1.1)	2.8 (1.9)
3m	-1.0 (-0.5)	-2.0 (-2.1)	-1.8 (-0.9)	-0.4 (-0.3)	1.6 (1.0)	2.1 (1.6)
6m	-0.7 (-0.4)	-1.0 (-1.2)	-0.6 (-0.3)	-0.1 (-0.1)	0.5 (0.5)	2.2 (1.7)
1y	0.4 (0.5)	-0.8 (-0.8)	0.1 (0.1)	-0.9 (-1.0)	1.0 (0.9)	1.8 (1.3)
2y	0.3 (0.4)	0.7 (0.6)	-0.2 (-0.1)	0.1 (0.1)	0.7 (0.7)	-0.9 (-1.2)
3y	1.2 (1.1)	0.5 (0.5)	0.2 (0.2)	-0.3 (-0.4)	0.3 (0.2)	-1.2 (-1.7)
5y	-0.4 (-0.2)	0.9 (0.7)	0.6 (0.5)	-0.3 (-0.5)	-0.8 (-0.6)	-0.8 (-1.1)

Close of 5/8/19. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 58. Implied correlations of 2s10s options



Note: The chart shows implied correlation of options on the 2s10s curve with different expirations. Source: Citi Research

Figure 59. Implied/realized ratios

implied/1m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.14 (-0.6)	1.17 (-0.5)	1.10 (-0.4)	1.35 (-0.2)	1.22 (-0.1)	1.26 (-0.2)
3m	1.24 (-0.6)	1.24 (-0.5)	1.10 (-0.4)	1.32 (-0.4)	1.16 (-0.3)	1.21 (-0.5)
6m	1.41 (-0.3)	1.34 (-0.4)	1.10 (-0.4)	1.31 (-0.5)	1.11 (-0.5)	1.25 (-0.5)
1y	1.65 (0.2)	1.41 (-0.3)	1.15 (-0.5)	1.34 (-0.7)	1.19 (-0.6)	1.29 (-0.6)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.30 (1.0)	1.25 (0.4)	1.16 (0.5)	1.37 (0.2)	1.28 (0.5)	1.27 (0.1)
3m	1.39 (0.7)	1.30 (0.1)	1.14 (0.1)	1.35 (0.0)	1.23 (0.2)	1.24 (-0.2)
6m	1.49 (0.7)	1.36 (0.1)	1.13 (-0.2)	1.37 (-0.1)	1.17 (-0.2)	1.27 (-0.3)
1y	1.56 (0.5)	1.36 (-0.2)	1.15 (-0.4)	1.40 (-0.3)	1.25 (-0.3)	1.31 (-0.5)

Close of 4/5/19. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Eurodollar convexity adjustments

We calculate ED/FRA convexity adjustment for 1y ED packs and compare them to model values calibrated to cap/floor volatilities.

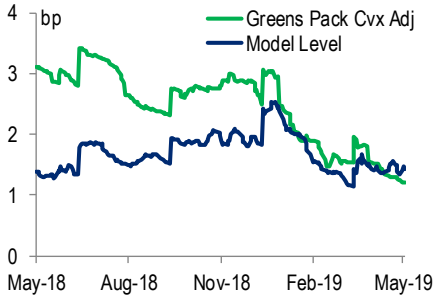
- Convexity adjustments have moved marginally lower over the past week. Golds Eurodollar convexity adjustments now trade only slightly rich to the model, while Blues convexities look cheap to the model (Figure 60).
- *Note: Futures trading involves substantial risk of loss*

Figure 60. Convexity adjustments for 1y Eurodollar packs (vs CME swaps)

	convexity adjustement				vs the model				implied vs realized vol			
ED	Cvx Adj	1 Week	3m	1Y	Vs Model	3m	1Y	3m Roll	Implied	Realized	Implied/	Cap vol
Pack	(bp)	Chg (bp)	Z-Score	Z-Score	(bp)	Z-Score	Z-Score	(short cvx, bp)	Vol	Vol	Realized	Impl/Rlzd
M9-H0	0.01	0.00	-1.01	**-0.86**	-0.03	-0.99	**-0.92**	0.03	30.6	53.8	0.6	0.8
U9-M0	0.08	0.00	-1.07	**-1.58**	-0.02	-1.25	-1.76	0.06	49.4	60.1	0.8	0.8
Z9-U0	0.17	-0.01	-1.34	-1.67	**-0.02**	-1.39	-1.85	0.10	57.0	66.0	0.9	0.8
H0-Z0	0.30	-0.01	-1.52	-1.75	**-0.02**	-1.45	-1.90	0.12	60.7	69.7	0.9	0.8
M0-H1	0.43	-0.02	-1.67	-1.84	-0.05	-1.46	-1.90	0.14	61.5	71.4	0.9	0.8
U0-M1	0.58	-0.03	-1.73	-1.89	-0.09	-1.52	-1.86	0.15	61.4	71.3	0.9	0.9
Z0-U1	0.75	-0.05	-1.72	-1.95	-0.14	-1.60	-1.82	0.17	61.2	70.1	0.9	0.9
H1-Z1	0.95	-0.06	-1.71	-2.00	-0.19	-1.66	-1.80	0.20	61.2	68.3	0.9	0.9
M1-H2	1.21	-0.08	-1.72	-2.03	-0.21	-1.71	-1.77	0.26	62.2	66.0	0.9	1.0
U1-M2	1.58	-0.09	-1.46	-1.85	-0.15	-1.43	-1.56	0.37	64.8	64.0	1.0	1.0
Z1-U2	1.83	-0.12	**-0.77**	-1.77	-0.26	**-0.71**	-1.43	0.25	63.8	62.2	1.0	1.1
H2-Z2	2.12	-0.19	**-0.48**	**-1.65**	-0.35	**-0.41**	-1.19	0.29	63.3	60.6	1.0	1.1
M2-H3	2.24	-0.32	**-0.69**	-1.76	-0.64	**-0.63**	-1.26	0.11	60.4	59.6	1.0	1.1
U2-M3	2.67	-0.39	-0.98	-1.79	-0.66	-0.91	-1.16	0.43	61.6	58.8	1.0	1.1
Z2-U3	3.21	-0.46	-1.61	-1.92	-0.59	-1.49	-1.16	**0.54**	63.3	58.5	**1.1**	1.1
H3-Z3	3.98	-0.41	-1.98	-1.95	-0.34	-1.44	**-1.10**	**0.77**	66.4	58.4	**1.1**	1.2
M3-H4	4.93	-0.22	-1.69	-1.91	**0.04**	-1.06	**-0.99**	**0.95**	69.7	58.2	**1.2**	1.2

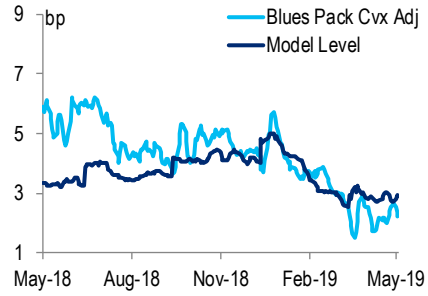
Note: Close of 5/8/19. Convexity adjustments for 1y ED packs are computed as the spread between the pack's rate (the average of 4 ED rates in the pack) and matched-maturity forward 1y CME swap rate. The model for convexity adjustment is the Ho-Lee model calibrated to cap/floor vols. Implied vol is calculated by matching the model to the observed convexity adjustment. Realized vol is 3m realized vol of the corresponding pack. For each valuation metric, we mark three best short convexity trades in **bold**. Note that these are the most attractive in the chart but not necessarily trades that we would recommend at this point in time. Source: Citi Research

Figure 61. Greens pack (rolling) vs cap/floor-based model



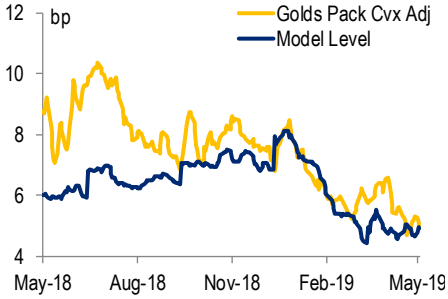
The chart shows the convexity adjustment of the Greens pack vs the (Ho-Lee) model. Source: Citi Research

Figure 62. Blues pack (rolling) vs cap/floor-based model



The chart shows the convexity adjustment of the Blues pack vs the (Ho-Lee) model. Source: Citi Research

Figure 63. Golds pack (rolling) vs cap/floor-based model



The chart shows the convexity adjustment of the Golds pack vs the (Ho-Lee) model. Source: Citi Research

Conditional curves

We analyze conditional curve trades based on the entry point, implied/realized betas, and carry.

- We still like 3y 5s30s conditional bear steepeners as recommended in US Rates Vol Lab: CCAR and bear steepeners. The zero-cost strike spread is about 8bp (pricing as of 3pm on 5/8/2019).

Prepared for Christopher Lee

Figure 64. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.5	-13.5	-12.3	-0.6	-6.0	0.7	-0.5	7	0.38	0.41	0.03	-1.2	-6.3	-7.4	-6.4	1.2	6.3	7.4	6.4
3m 1y3y	100/34.0	-20.0	-16.8	-0.3	-9.0	1.3	-0.4	6	0.45	0.51	0.06	-3.2	-7.8	-11.0	-6.4	3.2	7.8	11.0	6.4
3m 1y5y	100/20.8	-20.6	-15.2	-0.1	-6.6	3.0	-0.3	5	0.47	0.52	0.05	-5.4	-8.7	-14.0	-6.4	5.4	8.7	14.0	6.4
3m 2y5y	100/41.2	-7.1	-3.0	0.8	-1.4	2.0	0.2	4	0.08	0.10	0.03	-4.2	-1.6	-5.8	-5.3	4.2	1.6	5.8	5.3
3m 2y10y	100/21.7	7.2	12.7	0.6	13.2	2.2	0.0	7	0.02	0.08	0.06	-5.5	-0.4	-6.0	-5.3	5.5	0.4	6.0	5.3
3m 2y30y	100/9.1	25.1	30.2	0.5	28.2	2.2	0.1	10	-0.12	-0.05	0.08	-5.2	2.1	-3.1	-3.2	5.2	-2.1	3.1	3.2
3m 3y5y	100/61.2	-0.6	1.6	0.8	1.9	1.4	0.6	6	0.02	0.01	-0.01	-2.2	-0.3	-2.5	-2.6	2.2	0.3	2.5	2.6
3m 3y10y	100/32.3	13.7	17.3	0.5	16.5	1.7	0.1	11	-0.04	-0.02	0.02	-3.5	0.8	-2.8	-2.4	3.5	-0.8	2.8	2.4
3m 3y30y	100/13.5	31.6	34.8	0.4	31.4	1.6	0.1	12	-0.19	-0.14	0.04	-3.2	3.4	0.2	0.0	3.2	-3.4	-0.2	0.0
3m 5y10y	100/52.7	14.3	15.7	0.2	14.5	0.3	-0.3	15	-0.06	-0.03	0.03	-1.3	1.1	-0.2	0.0	1.3	-1.1	0.2	0.0
3m 5y30y	100/22.1	32.2	33.2	0.3	29.3	0.1	-0.1	15	-0.20	-0.15	0.05	-1.0	3.9	2.9	0.0	1.0	-3.9	-2.9	0.0
3m 10y30y	100/41.8	17.9	17.5	0.3	14.7	-0.1	0.0	16	-0.14	-0.12	0.02	0.4	2.7	3.1	0.0	-0.4	-2.7	-3.1	0.0
6m 1y2y	100/50.5	-13.5	-9.8	-0.2	-4.7	1.2	-0.3	8	0.20	0.28	0.08	-3.6	-5.1	-8.8	-1.7	3.6	5.1	8.8	1.7
6m 1y3y	100/33.9	-20.0	-12.4	0.2	-5.9	1.9	-0.3	7	0.24	0.35	0.10	-7.6	-6.5	-14.1	-2.8	7.6	6.5	14.1	2.8
6m 1y5y	100/20.8	-20.6	-8.8	0.7	-1.8	4.3	-0.1	6	0.25	0.35	0.10	-11.8	-7.0	-18.8	-3.7	11.8	7.0	18.8	3.7
6m 2y5y	100/41.2	-7.1	1.0	1.0	2.4	2.9	0.5	6	0.04	0.06	0.01	-8.2	-1.3	-9.5	-2.4	8.2	1.3	9.5	2.4
6m 2y10y	100/21.7	7.2	17.9	0.7	17.6	3.1	0.2	10	-0.01	0.02	0.03	-10.7	0.3	-10.4	-2.7	10.7	-0.3	10.4	2.7
6m 2y30y	100/9.1	25.1	34.9	0.6	31.0	2.8	0.1	11	-0.16	-0.12	0.04	-9.8	3.9	-5.9	-1.6	9.8	-3.9	5.9	1.6
6m 3y5y	100/61.3	-0.6	3.6	0.9	3.8	2.2	0.8	9	0.01	0.00	-0.01	-4.2	-0.2	-4.4	-1.1	4.2	0.2	4.4	1.1
6m 3y10y	100/32.4	13.7	20.4	0.6	19.1	2.4	0.2	13	-0.05	-0.04	0.01	-6.7	1.4	-5.3	-1.4	6.7	-1.4	5.3	1.4
6m 3y30y	100/13.5	31.6	37.4	0.5	32.3	2.2	0.0	13	-0.20	-0.18	0.02	-5.8	5.1	-0.7	-0.4	5.8	-5.1	0.7	0.4
6m 5y10y	100/52.8	14.3	16.8	0.2	15.2	0.2	-0.4	16	-0.05	-0.04	0.02	-2.5	1.6	-0.9	-0.4	2.5	-1.6	0.9	0.4
6m 5y30y	100/22.1	32.2	33.8	0.3	28.2	-0.1	-0.3	15	-0.20	-0.17	0.03	-1.6	5.6	4.0	0.5	1.6	-5.6	-4.0	-0.5
6m 10y30y	100/41.8	17.9	17.0	0.3	13.0	-0.3	-0.1	14	-0.14	-0.13	0.01	0.9	4.0	4.9	0.8	-0.9	-4.0	-4.9	-0.8
1y 1y2y	100/50.5	-13.5	-3.2	1.4	-0.2	1.3	0.3	11	0.07	0.13	0.06	-10.3	-3.0	-13.2	-1.8	10.3	3.0	13.2	1.8
1y 1y3y	100/34.0	-20.0	-2.0	1.2	0.9	2.8	0.9	9	0.07	0.16	0.09	-18.0	-2.9	-20.9	-2.5	18.0	2.9	20.9	2.5
1y 1y5y	100/20.8	-20.6	4.8	1.1	6.6	3.0	0.5	8	0.04	0.14	0.10	-25.4	-1.8	-27.2	-3.0	25.4	1.8	27.2	3.0
1y 2y5y	100/41.2	-7.1	8.0	0.8	6.7	1.7	0.6	8	-0.03	0.01	0.04	-15.1	1.3	-13.9	-1.2	15.1	-1.3	13.9	1.2
1y 2y10y	100/21.8	7.2	26.6	0.6	23.1	1.6	0.0	13	-0.09	-0.04	0.04	-19.4	3.5	-15.9	-1.3	19.4	-3.5	15.9	1.3
1y 2y30y	100/9.1	25.1	42.3	0.5	33.3	1.0	-0.1	12	-0.23	-0.18	0.05	-17.3	9.1	-8.2	-0.4	17.3	-9.1	8.2	0.4
1y 3y5y	100/61.2	-0.6	6.8	0.8	5.7	0.2	-0.3	12	-0.03	-0.02	0.01	-7.4	1.2	-6.3	-0.5	7.4	-1.2	6.3	0.5
1y 3y10y	100/32.3	13.7	25.4	0.4	21.9	0.1	-0.4	15	-0.08	-0.07	0.01	-11.7	3.5	-8.2	-0.7	11.7	-3.5	8.2	0.7
1y 3y30y	100/13.5	31.6	41.2	0.4	32.1	-0.6	-0.4	13	-0.23	-0.20	0.02	-9.6	9.1	-0.5	0.1	9.6	-9.1	0.5	-0.1
1y 5y10y	100/52.8	14.3	18.6	0.1	16.3	-0.1	-0.5	18	-0.05	-0.05	0.00	-4.2	2.3	-1.9	-0.2	4.2	-2.3	1.9	0.2
1y 5y30y	100/22.1	32.2	34.3	0.3	26.3	-0.8	-0.4	13	-0.20	-0.19	0.01	-2.1	8.0	5.9	0.5	2.1	-8.0	-5.9	-0.5
1y 10y30y	100/41.9	17.9	15.8	0.4	9.9	-0.7	-0.3	10	-0.14	-0.13	0.01	2.1	5.8	7.9	0.7	-2.1	-5.8	-7.9	-0.7
2y 1y2y	100/50.5	-13.5	3.3	0.6	3.5	-0.9	-1.3	15	0.00	0.09	0.09	-16.8	-0.1	-16.9	-0.1	16.8	0.1	16.9	0.1
2y 1y3y	100/34.0	-20.0	7.8	0.6	7.0	0.4	-0.2	12	-0.01	0.13	0.15	-27.8	0.8	-27.0	-0.1	27.8	-0.8	27.0	0.1
2y 1y5y	100/20.8	-20.6	16.9	0.6	14.4	0.0	-0.6	11	-0.04	0.13	0.17	-37.5	2.6	-35.0	-0.2	37.5	-2.6	35.0	0.2
2y 2y5y	100/41.3	-7.1	13.6	0.5	10.9	1.0	0.2	12	-0.04	0.03	0.07	-20.7	2.7	-18.0	-0.1	20.7	-2.7	18.0	0.1
2y 2y10y	100/21.8	7.2	32.4	0.1	25.8	0.0	-0.5	14	-0.10	-0.02	0.08	-25.2	6.6	-18.6	0.2	25.2	-6.6	18.6	-0.2
2y 2y30y	100/9.2	25.1	44.0	0.3	29.0	-0.6	-0.5	10	-0.24	-0.15	0.10	-18.9	14.9	-4.0	1.1	18.9	-14.9	4.0	-1.1
2y 3y5y	100/61.3	-0.6	9.1	0.3	7.3	-0.4	-0.8	14	-0.03	-0.01	0.02	-9.7	1.8	-7.9	-0.1	9.7	-1.8	7.9	0.1
2y 3y10y	100/32.4	13.7	27.9	-0.1	22.2	-1.4	-1.0	15	-0.09	-0.06	0.03	-14.1	5.6	-8.5	0.3	14.1	-5.6	8.5	-0.3
2y 3y30y	100/13.6	31.6	39.5	0.2	25.4	-2.1	-0.8	10	-0.23	-0.19	0.04	-7.9	14.0	6.2	1.1	7.9	-14.0	-6.2	-1.1
2y 5y10y	100/52.9	14.3	18.8	-0.4	14.9	-0.9	-1.0	16	-0.06	-0.05	0.01	-4.4	3.9	-0.6	0.3	4.4	-3.9	0.6	-0.3
2y 5y30y	100/22.2	32.2	30.4	0.1	18.1	-1.6	-0.7	7	-0.20	-0.18	0.02	1.8	12.3	14.1	1.1	-1.8	-12.3	-14.1	-1.1
2y 10y30y	100/41.9	17.9	11.6	0.4	3.3	-0.6	-0.3	5	-0.14	-0.12	0.01	6.3	8.3	14.5	0.8	-6.3	-8.3	-14.5	-0.8
3y 1y2y	100/50.5	-13.5	5.1	0.6	5.1	0.3	-0.1	17	0.00	0.15	0.15	-18.6	0.1	-18.5	-0.2	18.6	-0.1	18.5	0.2
3y 1y3y	100/34.0	-20.0	10.1	0.6	8.1	-0.6	-0.7	12	-0.02	0.20	0.23	-30.1	2.0	-28.1	0.1	30.1	-2.0	28.1	-0.1
3y 1y5y	100/20.9	-20.6	19.0	0.2	14.9	-0.7	-0.9	11	-0.05	0.20	0.25	-39.6	4.1	-35.5	0.2	39.6	-4.1	35.5	-0.2
3y 2y5y	100/41.3	-7.1	13.9	-0.2	9.8	-1.0	-0.7	11	-0.05	0.05	0.10	-21.1	4.1	-16.9	0.4	21.1	-4.1	16.9	-0.4
3y 2y10y	100/21.9	7.2	30.7	-0.4	21.4	-2.1	-0.9	12	-0.11	0.00	0.11	-23.5	9.2	-14.2	0.8	23.5	-9.2	14.2	-0.8
3y 2y30y	100/9.2	25.1	37.5	0.1	17.6	-3.1	-0.8	5	-0.25	-0.11	0.14	-12.4	19.9	7.5	1.5	12.4	-19.9	-7.5	-1.5
3y 3y5y	100/61.5	-0.6	8.9	-0.4	6.8	-0.1	-0.6	14	-0.03	-0.01	0.02	-9.5	2.2	-7.4	0.2	9.5	-2.2	7.4	-0.2
3y 3y10y	100/32.6	13.7	25.7	-0.5	18.4	-1.2	-0.9	13	-0.09	-0.05	0.04	-12.0	7.3	-4.7	0.6	12.0	-7.3	4.7	-0.6
3y 3y30y	100/13.7	31.6	32.5	0.0	14.6	-2.2	-0.8	4	-0.23	-0.17	0.06	-0.9	18.0	17.0	1.3	0.9	-18.0	-17.0	-1.3
3y 5y10y	100/53.0	14.3	16.7	-0.5	11.7	-1.1	-1.0	11	-0.06	-0.05	0.02	-2.4	5.0	2.6	0.4	2.4	-5.0	-2.6	-0.4
3y 5y30y	100/22.2	32.2	23.6	0.1	8.0	-1.9	-0.7	3	-0.20	-0.16	0.04	8.6	15.6	24.2	1.2	-8.6	-15.6	-24.2	-1.2
3y 10y30y	100/41.9	17.9	6.8	0.5	-3.4	-0.8	-0.2	1	-0.13	-0.11	0.02	11.0	10.3	21.3	0.8	-11.0	-10.3	-21.3	-0.8

Note: Close of 5/8/19. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bear steepeners in red, and three best bear flatteners in green. Source: Citi Research

Prepared for Christopher Lee

Figure 65. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.5	-13.5	-12.3	-0.6	-21.3	-1.6	-1.0	2	0.38	0.41	0.03	-1.2	9.0	7.9	0.0	1.2	-9.0	-7.9	0.0
3m 1y3y	100/34.0	-20.0	-16.8	-0.3	-27.6	-0.1	-0.4	1	0.45	0.51	0.06	-3.2	10.8	7.7	0.0	3.2	-10.8	-7.7	0.0
3m 1y5y	100/20.8	-20.6	-15.2	-0.1	-25.9	1.2	0.2	1	0.47	0.52	0.05	-5.4	10.7	5.3	0.0	5.4	-10.7	-5.3	0.0
3m 2y5y	100/41.2	-7.1	-3.0	0.8	-4.8	2.3	1.0	3	0.08	0.10	0.03	-4.2	1.9	-2.3	0.0	4.2	-1.9	2.3	0.0
3m 2y10y	100/21.7	7.2	12.7	0.6	12.2	3.0	0.9	7	0.02	0.08	0.06	-5.5	0.5	-5.1	0.0	5.5	-0.5	5.1	0.0
3m 2y30y	100/9.1	25.1	30.2	0.5	33.0	3.9	0.9	12	-0.12	-0.05	0.08	-5.2	-2.8	-8.0	-0.1	5.2	2.8	8.0	0.1
3m 3y5y	100/61.2	-0.6	1.6	0.8	1.2	1.0	0.8	4	0.02	0.01	-0.01	-2.2	0.4	-1.8	0.0	2.2	-0.4	1.8	0.0
3m 3y10y	100/32.3	13.7	17.3	0.5	18.3	1.7	0.7	13	-0.04	-0.02	0.02	-3.5	-1.0	-4.5	-0.5	3.5	1.0	4.5	0.5
3m 3y30y	100/13.5	31.6	34.8	0.4	39.2	2.6	0.7	16	-0.19	-0.14	0.04	-3.2	-4.4	-7.6	-0.1	3.2	4.4	7.6	0.1
3m 5y10y	100/52.7	14.3	15.7	0.2	17.0	0.7	0.6	18	-0.06	-0.03	0.03	-1.3	-1.4	-2.7	-0.5	1.3	1.4	2.7	0.5
3m 5y30y	100/22.1	32.2	33.2	0.3	37.8	1.5	0.6	21	-0.20	-0.15	0.05	-1.0	-4.7	-5.6	-0.1	1.0	4.7	5.6	0.1
3m 10y30y	100/41.8	17.9	17.5	0.3	20.5	0.8	0.6	26	-0.14	-0.12	0.02	0.4	-3.0	-2.7	-0.1	-0.4	3.0	2.7	0.1
6m 1y2y	100/50.5	-13.5	-9.8	-0.2	-17.2	-0.4	0.2	3	0.20	0.28	0.08	-3.6	7.3	3.7	-1.6	3.6	-7.3	-3.7	1.6
6m 1y3y	100/33.9	-20.0	-12.4	0.2	-21.0	1.2	0.3	3	0.24	0.35	0.10	-7.6	8.6	1.0	-2.3	7.6	-8.6	-1.0	2.3
6m 1y5y	100/20.8	-20.6	-8.8	0.7	-17.1	1.1	1.0	3	0.25	0.35	0.10	-11.8	8.4	-3.5	-3.0	11.8	-8.4	3.5	3.0
6m 2y5y	100/41.2	-7.1	1.0	1.0	-0.4	1.6	1.1	4	0.04	0.06	0.01	-8.2	1.4	-6.8	-1.8	8.2	-1.4	6.8	1.8
6m 2y10y	100/21.7	7.2	17.9	0.7	18.5	2.3	0.9	10	-0.01	0.02	0.03	-10.7	-0.6	-11.3	-2.5	10.7	0.6	11.3	2.5
6m 2y30y	100/9.1	25.1	34.9	0.6	40.6	3.3	0.9	15	-0.16	-0.12	0.04	-9.8	-5.7	-15.5	-2.6	9.8	5.7	15.5	2.6
6m 3y5y	100/61.3	-0.6	3.6	0.9	3.4	0.1	0.8	8	0.01	0.00	-0.01	-4.2	0.2	-4.0	-0.9	4.2	-0.2	4.0	0.9
6m 3y10y	100/32.4	13.7	20.4	0.6	22.2	0.7	0.7	15	-0.05	-0.04	0.01	-6.7	-1.8	-8.5	-1.7	6.7	1.8	8.5	1.7
6m 3y30y	100/13.5	31.6	37.4	0.5	44.4	1.7	0.8	18	-0.20	-0.18	0.02	-5.8	-7.0	-12.8	-1.8	5.8	7.0	12.8	1.8
6m 5y10y	100/52.8	14.3	16.8	0.2	18.8	0.7	0.6	20	-0.05	-0.04	0.02	-2.5	-1.9	-4.4	-0.8	2.5	1.9	4.4	0.8
6m 5y30y	100/22.1	32.2	33.8	0.3	40.7	1.8	0.7	23	-0.20	-0.17	0.03	-1.6	-6.8	-8.5	-1.1	1.6	6.8	8.5	1.1
6m 10y30y	100/41.8	17.9	17.0	0.3	21.5	1.0	0.7	28	-0.14	-0.13	0.01	0.9	-4.5	-3.6	-0.4	-0.9	4.5	3.6	0.4
1y 1y2y	100/50.5	-13.5	-3.2	1.4	-7.1	1.4	1.2	6	0.07	0.13	0.06	-10.3	3.9	-6.4	-1.8	10.3	-3.9	6.4	1.8
1y 1y3y	100/34.0	-20.0	-2.0	1.2	-5.7	1.0	1.0	7	0.07	0.16	0.09	-18.0	3.7	-14.3	-2.8	18.0	-3.7	14.3	2.8
1y 1y5y	100/20.8	-20.6	4.8	1.1	2.8	2.9	1.0	7	0.04	0.14	0.10	-25.4	2.0	-23.4	-3.7	25.4	-2.0	23.4	3.7
1y 2y5y	100/41.2	-7.1	8.0	0.8	9.7	1.6	0.9	11	-0.03	0.01	0.04	-15.1	-1.7	-16.9	-1.9	15.1	1.7	16.9	1.9
1y 2y10y	100/21.8	7.2	26.6	0.6	31.3	2.2	0.8	16	-0.09	-0.04	0.04	-19.4	-4.8	-24.1	-2.4	19.4	4.8	24.1	2.4
1y 2y30y	100/9.1	25.1	42.3	0.5	54.9	4.0	0.9	19	-0.23	-0.18	0.05	-17.3	-12.5	-29.8	-2.2	17.3	12.5	29.8	2.2
1y 3y5y	100/61.2	-0.6	6.8	0.8	8.4	1.9	1.1	16	-0.03	-0.02	0.01	-7.4	-1.5	-9.0	-0.9	7.4	1.5	9.0	0.9
1y 3y10y	100/32.3	13.7	25.4	0.4	29.9	2.6	0.9	19	-0.08	-0.07	0.01	-11.7	-4.5	-16.2	-1.4	11.7	4.5	16.2	1.4
1y 3y30y	100/13.5	31.6	41.2	0.4	53.3	4.4	0.9	22	-0.23	-0.20	0.02	-9.6	-12.1	-21.7	-1.3	9.6	12.1	21.7	1.3
1y 5y10y	100/52.8	14.3	18.6	0.1	21.4	0.6	0.6	23	-0.05	-0.05	0.00	-4.2	-2.8	-7.1	-0.5	4.2	2.8	7.1	0.5
1y 5y30y	100/22.1	32.2	34.3	0.3	44.3	2.3	0.8	24	-0.20	-0.19	0.01	-2.1	-10.0	-12.1	-0.6	2.1	10.0	12.1	0.6
1y 10y30y	100/41.9	17.9	15.8	0.4	22.2	1.5	0.8	29	-0.14	-0.13	0.01	2.1	-6.5	-4.4	-0.1	-2.1	6.5	4.4	0.1
2y 1y2y	100/50.5	-13.5	3.3	0.6	3.2	2.0	1.3	14	0.00	0.09	0.09	-16.8	0.1	-16.7	-0.9	16.8	-0.1	16.7	0.9
2y 1y3y	100/34.0	-20.0	7.8	0.6	8.9	1.8	0.9	13	-0.01	0.13	0.15	-27.8	-1.1	-28.9	-1.4	27.8	1.1	28.9	1.4
2y 1y5y	100/20.8	-20.6	16.9	0.6	20.1	2.1	0.9	14	-0.04	0.13	0.17	-37.5	-3.2	-40.7	-1.4	37.5	3.2	40.7	1.4
2y 2y5y	100/41.3	-7.1	13.6	0.5	16.9	0.1	0.5	17	-0.04	0.03	0.07	-20.7	-3.3	-24.0	-0.5	20.7	3.3	24.0	0.5
2y 2y10y	100/21.8	7.2	32.4	0.1	40.3	0.9	0.5	20	-0.10	-0.02	0.08	-25.2	-8.0	-33.1	-0.6	25.2	8.0	33.1	0.6
2y 2y30y	100/9.2	25.1	44.0	0.3	62.5	2.3	0.7	22	-0.24	-0.15	0.10	-18.9	-18.5	-37.4	-0.2	18.9	18.5	37.4	0.2
2y 3y5y	100/61.3	-0.6	9.1	0.3	11.1	0.3	0.9	20	-0.03	-0.01	0.02	-9.7	-2.0	-11.7	0.0	9.7	2.0	11.7	0.0
2y 3y10y	100/32.4	13.7	27.9	-0.1	34.4	1.2	0.7	22	-0.09	-0.06	0.03	-14.1	-6.6	-20.7	-0.1	14.1	6.6	20.7	0.1
2y 3y30y	100/13.6	31.6	39.5	0.2	56.3	2.6	0.8	23	-0.23	-0.19	0.04	-7.9	-16.8	-24.7	0.2	7.9	16.8	24.7	-0.2
2y 5y10y	100/52.9	14.3	18.8	-0.4	23.1	0.9	0.4	24	-0.06	-0.05	0.01	-4.4	-4.4	-8.8	-0.1	4.4	4.4	8.8	0.1
2y 5y30y	100/22.2	32.2	30.4	0.1	44.5	2.3	0.7	25	-0.20	-0.18	0.02	1.8	-14.1	-12.3	0.2	-1.8	14.1	12.3	-0.2
2y 10y30y	100/41.9	17.9	11.6	0.4	20.8	1.2	0.9	27	-0.14	-0.12	0.01	6.3	-9.1	-2.9	0.2	-6.3	9.1	2.9	-0.2
3y 1y2y	100/50.5	-13.5	5.1	0.6	5.2	1.5	0.9	17	0.00	0.15	0.15	-18.6	-0.1	-18.7	-0.3	18.6	0.1	18.7	0.3
3y 1y3y	100/34.0	-20.0	10.1	0.6	12.4	2.0	1.1	16	-0.02	0.20	0.23	-30.1	-2.3	-32.4	-0.2	30.1	2.3	32.4	0.2
3y 1y5y	100/20.9	-20.6	19.0	0.2	23.9	1.7	0.9	16	-0.05	0.20	0.25	-39.6	-4.9	-44.5	-0.3	39.6	4.9	44.5	0.3
3y 2y5y	100/41.3	-7.1	13.9	-0.2	18.6	0.2	0.7	19	-0.05	0.05	0.10	-21.1	-4.7	-25.8	0.0	21.1	4.7	25.8	0.0
3y 2y10y	100/21.9	7.2	30.7	-0.4	41.3	1.0	0.6	21	-0.11	0.00	0.11	-23.5	-10.6	-34.1	0.0	23.5	10.6	34.1	0.0
3y 2y30y	100/9.2	25.1	37.5	0.1	60.8	2.7	0.9	21	-0.25	-0.11	0.14	-12.4	-23.3	-35.7	0.3	12.4	23.3	35.7	-0.3
3y 3y5y	100/61.5	-0.6	8.9	-0.4	11.4	-0.3	0.3	20	-0.03	-0.01	0.02	-9.5	-2.5	-12.0	-0.1	9.5	2.5	12.0	0.1
3y 3y10y	100/32.6	13.7	25.7	-0.5	33.9	0.4	0.4	22	-0.09	-0.05	0.04	-12.0	-8.2	-20.2	0.0	12.0	8.2	20.2	0.0
3y 3y30y	100/13.7	31.6	32.5	0.0	53.0	2.2	0.8	22	-0.23	-0.17	0.06	-0.9	-20.5	-21.4	0.3	0.9	20.5	21.4	-0.3
3y 5y10y	100/53.0	14.3	16.7	-0.5	22.3	0.7	0.4	24	-0.06	-0.05	0.02	-2.4	-5.6	-8.0	0.1	2.4	5.6	8.0	-0.1
3y 5y30y	100/22.2	32.2	23.6	0.1	41.1	2.5	0.9	23	-0.20	-0.16	0.04	8.6	-17.5	-8.9	0.3	-8.6	17.5	8.9	-0.3
3y 10y30y	100/41.9	17.9	6.8	0.5	18.0	1.6	1.0	22	-0.13	-0.11	0.02	11.0	-11.1	-0.1	0.3	-11.0	11.1	0.1	-0.3

Note: Close of 5/8/19. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best bull steepeners in red, and three best bull flatteners in green. Source: Citi Research

Prepared for Christopher Lee

Conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

- Buying 3m7y payers on the 3m 3s7s15s payer fly could be an attractive way to position short in the belly of the curve, given the richness of the fly (Figure 66). The main risk is the de-correlation of the fly and rates.

Figure 66. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	147.1/100/30.6	-9.7	-9.2	-0.5	91	90	1.5	0.5	3.5	-5.7	0.4	-0.5	26	-0.5	-3.5	-4.0	-3.7	0.5	3.5	4.0	3.7
3m 1y3y7y	147.1/100/22.3	-12.2	-12.1	-0.6	89	90	1.6	0.5	3.7	-8.4	0.5	-0.6	48	-0.1	-3.7	-3.7	-3.3	0.1	3.7	3.7	3.3
3m 2y3y5y	74.3/100/30.6	-3.0	-3.1	-0.5	58	91	0.5	0.6	0.4	-2.6	-0.3	-0.7	77	0.1	-0.4	-0.3	-0.3	-0.1	0.4	0.3	0.3
3m 2y5y7y	121.3/100/36.5	-6.1	-4.4	0.7	48	77	1.3	0.6	1.0	-3.4	1.1	0.2	4	-1.7	-1.0	-2.6	-2.7	1.7	1.0	2.6	2.7
3m 2y5y10y	121.3/100/26.4	-10.7	-9.3	0.7	63	82	1.6	0.6	1.4	-8.0	1.3	0.4	11	-1.4	-1.4	-2.8	-2.7	1.4	1.4	2.8	2.7
3m 2y10y30y	230.0/100/20.9	-5.3	-2.4	0.3	60	70	1.1	0.4	1.6	-0.8	1.3	0.0	8	-3.0	-1.6	-4.5	-2.7	3.0	1.6	4.5	2.7
3m 3y5y7y	81.7/100/36.5	-2.8	-2.1	0.8	20	68	0.4	0.5	0.3	-1.8	0.7	0.5	6	-0.7	-0.3	-1.0	-1.0	0.7	0.3	1.0	1.0
3m 3y5y10y	81.7/100/26.4	-7.5	-7.0	0.7	47	82	0.8	0.5	0.7	-6.3	0.9	0.8	38	-0.4	-0.7	-1.2	-1.4	0.4	0.7	1.2	1.4
3m 3y7y10y	112.0/100/36.2	-2.5	-1.2	1.0	36	39	0.8	0.7	0.4	-0.8	0.9	1.0	3	-1.3	-0.4	-1.7	-1.6	1.3	0.4	1.7	1.6
3m 3y7y15y	112.0/100/25.5	-8.1	-6.8	0.9	64	77	1.0	0.6	0.9	-5.9	1.2	1.0	6	-1.3	-0.9	-2.2	-1.6	1.3	0.9	2.2	1.6
3m 3y10y30y	154.9/100/20.9	-2.1	-0.1	0.2	53	56	0.3	0.2	0.9	0.8	0.9	0.0	13	-2.0	-0.9	-2.9	-1.6	2.0	0.9	2.9	1.6
3m 5y10y15y	94.8/100/35.3	1.5	2.3	0.3	27	-58	0.0	0.0	-0.1	2.2	0.1	0.0	18	-0.7	0.1	-0.6	-0.5	0.7	-0.1	0.6	0.5
3m 5y10y30y	94.8/100/20.9	-1.8	-0.9	-0.3	63	77	-0.2	-0.2	0.7	-0.2	0.3	-0.3	28	-0.9	-0.7	-1.6	-0.5	0.9	0.7	1.6	0.5
3m 10y20y30y	89.1/100/37.3	7.2	7.1	0.0	-40	-75	0.2	0.2	-0.5	6.6	-0.3	-0.3	21	0.1	0.5	0.5	0.3	-0.1	-0.5	-0.5	-0.3
6m 1y3y5y	147.4/100/30.7	-9.7	-8.0	-0.3	81	88	1.6	0.6	3.0	-4.9	0.5	-0.5	26	-1.7	-3.0	-4.8	-0.9	1.7	3.0	4.8	0.9
6m 1y3y7y	147.4/100/22.4	-12.2	-11.3	-0.4	81	89	1.7	0.6	3.4	-7.9	0.5	-0.5	47	-0.9	-3.4	-4.3	-0.8	0.9	3.4	4.3	0.8
6m 2y3y5y	74.4/100/30.7	-3.0	-3.1	-0.5	42	91	0.4	0.6	0.4	-2.6	-0.5	-0.9	77	0.1	-0.4	-0.3	-0.1	-0.1	0.4	0.3	0.1
6m 2y5y7y	121.3/100/36.5	-6.1	-2.8	1.1	15	61	1.1	0.7	1.0	-1.8	1.3	0.5	5	-3.3	-1.0	-4.3	-1.1	3.3	1.0	4.3	1.1
6m 2y5y10y	121.3/100/26.4	-10.7	-7.9	1.1	43	79	1.5	0.7	1.5	-6.4	1.6	0.7	13	-2.9	-1.5	-4.3	-1.0	2.9	1.5	4.3	1.0
6m 2y10y30y	229.9/100/20.9	-5.3	0.4	0.6	39	42	0.7	0.3	1.8	2.3	1.5	0.2	11	-5.8	-1.8	-7.6	-1.8	5.8	1.8	7.6	1.8
6m 3y5y7y	81.5/100/36.5	-2.8	-1.5	1.0	8	51	0.4	0.6	0.4	-1.1	0.9	0.7	8	-1.3	-0.4	-1.7	-0.4	1.3	0.4	1.7	0.4
6m 3y5y10y	81.5/100/26.4	-7.5	-6.6	0.8	46	82	0.8	0.6	0.9	-5.7	1.2	1.0	40	-0.8	-0.9	-1.8	-0.4	0.8	0.9	1.8	0.4
6m 3y7y10y	111.8/100/36.2	-2.5	0.0	1.1	16	-25	0.7	0.7	0.2	0.2	1.1	1.3	5	-2.4	-0.2	-2.7	-0.6	2.4	0.2	2.7	0.6
6m 3y7y15y	111.8/100/25.5	-8.1	-5.5	1.0	55	71	0.9	0.7	1.1	-4.5	1.5	1.2	10	-2.6	-1.1	-3.6	-0.8	2.6	1.1	3.6	0.8
6m 3y10y30y	154.5/100/20.9	-2.1	1.7	0.3	40	14	0.1	0.1	1.2	2.9	1.2	0.2	17	-3.8	-1.2	-5.0	-1.2	3.8	1.2	5.0	1.2
6m 5y10y15y	94.7/100/35.3	1.5	2.9	0.3	14	-72	-0.1	-0.1	0.0	2.9	0.0	-0.2	20	-1.4	0.0	-1.4	-0.4	1.4	0.0	1.4	0.4
6m 5y10y30y	94.7/100/20.9	-1.8	-0.1	-0.4	59	72	-0.3	-0.3	1.1	1.0	0.3	-0.2	34	-1.7	-1.1	-2.8	-0.6	1.7	1.1	2.8	0.6
6m 10y20y30y	89.1/100/37.3	7.2	7.0	0.1	-46	-76	0.2	0.2	-0.9	6.1	-0.4	-0.8	19	0.1	0.9	1.1	0.3	-0.1	-0.9	-1.1	-0.3
1y 1y3y5y	147.1/100/30.6	-9.7	-4.4	1.6	16	80	1.4	0.9	2.0	-2.4	1.5	1.3	24	-5.3	-2.0	-7.3	-1.0	5.3	2.0	7.3	1.0
1y 1y3y7y	147.1/100/22.3	-12.2	-8.3	1.1	40	86	1.7	0.8	2.4	-5.9	1.5	0.8	42	-3.9	-2.4	-6.3	-0.8	3.9	2.4	6.3	0.8
1y 2y3y5y	74.3/100/30.6	-3.0	-2.8	-0.4	21	90	0.3	0.5	0.5	-2.3	0.8	2.3	68	-0.1	-0.5	-0.7	-0.1	0.1	0.5	0.7	0.1
1y 2y5y7y	121.3/100/36.5	-6.1	0.1	1.0	-21	-52	0.6	0.6	-0.3	-0.2	0.7	0.4	6	-6.2	0.3	-5.9	-0.5	6.2	-0.3	5.9	0.5
1y 2y5y10y	121.3/100/26.4	-10.7	-5.3	1.1	16	59	1.0	0.8	0.5	-4.8	1.2	1.1	14	-5.5	-0.5	-5.9	-0.5	5.5	0.5	5.9	0.5
1y 2y10y30y	229.7/100/20.9	-5.3	5.4	0.6	10	-62	0.0	0.0	0.9	6.3	1.1	0.3	16	-10.7	-0.9	-11.7	-1.1	10.7	0.9	11.7	1.1
1y 3y5y7y	81.7/100/36.5	-2.8	-0.5	1.0	-4	-41	0.3	0.6	-0.3	-0.7	-0.1	-0.9	12	-2.3	0.3	-2.1	-0.1	2.3	-0.3	2.1	0.1
1y 3y5y10y	81.7/100/26.4	-7.5	-5.9	1.0	53	80	0.7	0.7	0.5	-5.3	0.4	0.4	45	-1.6	-0.5	-2.1	-0.2	1.6	0.5	2.1	0.2
1y 3y7y10y	111.9/100/36.2	-2.5	1.9	1.0	-9	-82	0.3	0.4	-0.1	1.8	0.5	1.5	10	-4.4	0.1	-4.2	-0.4	4.4	-0.1	4.2	0.4
1y 3y7y15y	111.9/100/25.6	-8.1	-3.3	1.0	42	41	0.5	0.6	1.1	-2.2	0.9	1.3	20	-4.8	-1.1	-5.9	-0.6	4.8	1.1	5.9	0.6
1y 3y10y30y	154.6/100/20.9	-2.1	4.8	0.1	24	-60	-0.3	-0.3	1.0	5.8	0.2	-0.3	24	-6.9	-1.0	-7.9	-0.8	6.9	1.0	7.9	0.8
1y 5y10y15y	94.6/100/35.3	1.5	4.0	0.0	-5	-78	-0.3	-0.3	0.0	4.1	0.0	-0.4	25	-2.5	0.0	-2.5	-0.3	2.5	0.0	2.5	0.3
1y 5y10y30y	94.6/100/20.9	-1.8	1.4	-0.6	53	54	-0.5	-0.7	1.6	3.0	0.3	-0.1	43	-3.2	-1.6	-4.8	-0.5	3.2	1.6	4.8	0.5
1y 10y20y30y	89.0/100/37.3	7.2	6.8	0.1	-51	-77	0.2	0.2	-1.3	5.5	-0.4	-1.0	17	0.4	1.3	1.7	0.2	-0.4	-1.3	-1.7	-0.2

Note: Close of 5/8/19. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly payer trades in red**, and three **best short belly payer trades in green**. Source: Citi Research

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Figure 67. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly				costless strike					long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	147.1/100/30.6	-9.7	-9.2	-0.5	91	90	1.5	0.5	4.6	-13.8	0.0	-0.7	15	0.5	-4.6	-4.1	0.0	-0.5	4.6	4.1	0.0
3m 1y3y7y	147.1/100/22.3	-12.2	-12.1	-0.6	89	90	1.6	0.5	4.9	-17.0	0.0	-0.7	28	0.1	-4.9	-4.8	0.0	-0.1	4.9	4.8	0.0
3m 2y3y5y	74.3/100/30.6	-3.0	-3.1	-0.5	58	91	0.5	0.6	0.6	-3.6	0.3	-0.3	68	-0.1	-0.6	-0.7	0.0	0.1	0.6	0.7	0.0
3m 2y5y7y	121.3/100/36.5	-6.1	-4.4	0.7	48	77	1.3	0.6	1.1	-5.5	1.4	1.0	0	1.7	-1.1	0.5	0.0	-1.7	1.1	-0.5	0.0
3m 2y5y10y	121.3/100/26.4	-10.7	-9.3	0.7	63	82	1.6	0.6	1.6	-10.9	1.4	0.8	7	1.4	-1.6	-0.2	-0.2	-1.4	1.6	0.2	0.2
3m 2y10y30y	230.0/100/20.9	-5.3	-2.4	0.3	60	70	1.1	0.4	1.7	-4.1	1.3	0.6	2	3.0	-1.7	1.2	0.0	-3.0	1.7	-1.2	0.0
3m 3y5y7y	81.7/100/36.5	-2.8	-2.1	0.8	20	68	0.4	0.5	0.4	-2.5	0.6	0.7	2	0.7	-0.4	0.3	0.0	-0.7	0.4	-0.3	0.0
3m 3y5y10y	81.7/100/26.4	-7.5	-7.0	0.7	47	82	0.8	0.5	0.9	-7.9	0.6	0.3	30	0.4	-0.9	-0.4	-0.2	-0.4	0.9	0.4	0.2
3m 3y7y10y	112.0/100/36.2	-2.5	-1.2	1.0	36	39	0.8	0.7	0.4	-1.6	0.5	0.8	1	1.3	-0.4	0.8	-0.2	-1.3	0.4	-0.8	0.2
3m 3y7y15y	112.0/100/25.5	-8.1	-6.8	0.9	64	77	1.0	0.6	1.0	-7.8	0.6	0.7	3	1.3	-1.0	0.3	-0.2	-1.3	1.0	-0.3	0.2
3m 3y10y30y	154.9/100/20.9	-2.1	-0.1	0.2	53	56	0.3	0.2	1.0	-1.1	0.5	0.4	5	2.0	-1.0	1.0	0.0	-2.0	1.0	-1.0	0.0
3m 5y10y15y	94.8/100/35.3	1.5	2.3	0.3	27	-58	0.0	0.0	-0.1	2.3	0.1	0.7	16	0.7	0.1	0.8	0.4	-0.7	-0.1	-0.8	-0.4
3m 5y10y30y	94.8/100/20.9	-1.8	-0.9	-0.3	63	77	-0.2	-0.2	0.8	-1.7	0.0	-0.4	19	0.9	-0.8	0.1	0.0	-0.9	0.8	-0.1	0.0
3m 10y20y30y	89.1/100/37.3	7.2	7.1	0.0	-40	-75	0.2	0.2	-0.5	7.7	-0.2	0.3	23	-0.1	0.5	0.5	0.5	0.1	-0.5	-0.5	-0.5
6m 1y3y5y	147.4/100/30.7	-9.7	-8.0	-0.3	81	88	1.6	0.6	4.0	-11.9	0.9	0.2	12	1.7	-4.0	-2.2	0.7	-1.7	4.0	2.2	-0.7
6m 1y3y7y	147.4/100/22.4	-12.2	-11.3	-0.4	81	89	1.7	0.6	4.4	-15.7	1.0	-0.1	25	0.9	-4.4	-3.5	0.4	-0.9	4.4	3.5	-0.4
6m 2y3y5y	74.4/100/30.7	-3.0	-3.1	-0.5	42	91	0.4	0.6	0.5	-3.6	0.8	0.2	62	-0.1	-0.5	-0.6	0.0	0.1	0.5	0.6	0.0
6m 2y5y7y	121.3/100/36.5	-6.1	-2.8	1.1	15	61	1.1	0.7	1.1	-3.9	1.1	1.2	1	3.3	-1.1	2.2	0.6	-3.3	1.1	-2.2	-0.6
6m 2y5y10y	121.3/100/26.4	-10.7	-7.9	1.1	43	79	1.5	0.7	1.7	-9.6	1.1	1.2	7	2.9	-1.7	1.2	0.5	-2.9	1.7	-1.2	-0.5
6m 2y10y30y	229.9/100/20.9	-5.3	0.4	0.6	39	42	0.7	0.3	1.9	-1.5	0.9	0.9	3	5.8	-1.9	3.9	1.1	-5.8	1.9	-3.9	-1.1
6m 3y5y7y	81.5/100/36.5	-2.8	-1.5	1.0	8	51	0.4	0.6	0.5	-2.0	0.2	0.8	2	1.3	-0.5	0.8	0.2	-1.3	0.5	-0.8	-0.2
6m 3y5y10y	81.5/100/26.4	-7.5	-6.6	0.8	46	82	0.8	0.6	1.1	-7.7	0.2	0.3	31	0.8	-1.1	-0.2	0.1	-0.8	1.1	0.2	-0.1
6m 3y7y10y	111.8/100/36.2	-2.5	0.0	1.1	16	-25	0.7	0.7	0.2	-0.3	0.0	0.7	1	2.4	-0.2	2.2	0.6	-2.4	0.2	-2.2	-0.6
6m 3y7y15y	111.8/100/25.5	-8.1	-5.5	1.0	55	71	0.9	0.7	1.2	-6.7	0.1	0.6	4	2.6	-1.2	1.4	0.4	-2.6	1.2	-1.4	-0.4
6m 3y10y30y	154.5/100/20.9	-2.1	1.7	0.3	40	14	0.1	0.1	1.3	0.4	0.0	0.3	8	3.8	-1.3	2.5	0.6	-3.8	1.3	-2.5	-0.6
6m 5y10y15y	94.7/100/35.3	1.5	2.9	0.3	14	-72	-0.1	-0.1	0.0	2.9	0.0	0.7	17	1.4	0.0	1.4	0.3	-1.4	0.0	-1.4	-0.3
6m 5y10y30y	94.7/100/20.9	-1.8	-0.1	-0.4	59	72	-0.3	-0.3	1.2	-1.3	-0.1	-0.6	22	1.7	-1.2	0.5	0.2	-1.7	1.2	-0.5	-0.2
6m 10y20y30y	89.1/100/37.3	7.2	7.0	0.1	-46	-76	0.2	0.2	-1.0	8.0	-0.1	0.6	23	-0.1	1.0	0.9	0.2	0.1	-1.0	-0.9	-0.2
1y 1y3y5y	147.1/100/30.6	-9.7	-4.4	1.6	16	80	1.4	0.9	2.6	-7.0	-0.2	0.6	16	5.3	-2.6	2.7	0.9	-5.3	2.6	-2.7	-0.9
1y 1y3y7y	147.1/100/22.3	-12.2	-8.3	1.1	40	86	1.7	0.8	3.0	-11.4	0.2	0.7	29	3.9	-3.0	0.8	0.8	-3.9	3.0	-0.8	-0.8
1y 2y3y5y	74.3/100/30.6	-3.0	-2.8	-0.4	21	90	0.3	0.5	0.7	-3.5	-1.0	-1.6	69	0.1	-0.7	-0.6	0.1	-0.1	0.7	0.6	-0.1
1y 2y5y7y	121.3/100/36.5	-6.1	0.1	1.0	-21	-52	0.6	0.6	-0.4	0.5	0.6	1.1	4	6.2	0.4	6.6	0.9	-6.2	-0.4	-6.6	-0.9
1y 2y5y10y	121.3/100/26.4	-10.7	-5.3	1.1	16	59	1.0	0.8	0.6	-5.9	0.5	1.0	10	5.5	-0.6	4.9	0.7	-5.5	0.6	-4.9	-0.7
1y 2y10y30y	229.7/100/20.9	-5.3	5.4	0.6	10	-62	0.0	0.0	1.0	4.4	-0.1	0.5	9	10.7	-1.0	9.7	1.2	-10.7	1.0	-9.7	-1.2
1y 3y5y7y	81.7/100/36.5	-2.8	-0.5	1.0	-4	-41	0.3	0.6	-0.3	-0.1	1.0	1.6	6	2.3	0.3	2.7	0.4	-2.3	-0.3	-2.7	-0.4
1y 3y5y10y	81.7/100/26.4	-7.5	-5.9	1.0	53	80	0.7	0.7	0.7	-6.5	0.8	1.4	34	1.6	-0.7	1.0	0.2	-1.6	0.7	-1.0	-0.2
1y 3y7y10y	111.9/100/36.2	-2.5	1.9	1.0	-9	-82	0.3	0.4	-0.2	2.1	0.2	0.7	6	4.4	0.2	4.6	0.4	-4.4	-0.2	-4.6	-0.4
1y 3y7y15y	111.9/100/25.6	-8.1	-3.3	1.0	42	41	0.5	0.6	1.2	-4.6	0.2	0.6	8	4.8	-1.2	3.5	0.4	-4.8	1.2	-3.5	-0.4
1y 3y10y30y	154.6/100/20.9	-2.1	4.8	0.1	24	-60	-0.3	-0.3	1.1	3.8	0.2	0.4	15	6.9	-1.1	5.8	0.7	-6.9	1.1	-5.8	-0.7
1y 5y10y15y	94.6/100/35.3	1.5	4.0	0.0	-5	-78	-0.3	-0.3	0.1	4.0	-0.2	0.4	21	2.5	-0.1	2.4	0.2	-2.5	0.1	-2.4	-0.2
1y 5y10y30y	94.6/100/20.9	-1.8	1.4	-0.6	53	54	-0.5	-0.7	1.8	-0.4	-0.4	-1.0	28	3.2	-1.8	1.4	0.2	-3.2	1.8	-1.4	-0.2
1y 10y20y30y	89.0/100/37.3	7.2	6.8	0.1	-51	-77	0.2	0.2	-1.4	8.2	-0.1	0.8	22	-0.4	1.4	1.0	0.0	0.4	-1.4	-1.0	0.0

Note: Close of 5/8/19. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly receiver trades in red**, and three **best short belly receiver trades in green**. Source: Citi Research

Past trade performance

Please refer to the latest US Rates Weekly for the past 12 month performance of our trade portfolio (see US Rates Weekly: Leverage trap).

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Appendix A-1

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